Commutative Algebra

HHH

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Contents

Chapter I

§1 Ideals quotient

Definition 1.1. *If* $\mathfrak{a}, \mathfrak{b}$ *are ideals in a commutative ring* R*, their* ***ideal quotient*** *is*

$$(\mathfrak{a} : \mathfrak{b}) = \{x \in R : x\mathfrak{b} \subseteq \mathfrak{a}\}$$

which is an ideal. If $\mathfrak{b}$ *is a principal ideal* (x)*, we shall write* $(\mathfrak{a} : x)$ *in place of* $(\mathfrak{a} : (x))$*.*

Proposition 1.2. *Let* R *be a commutative ring. Then*

1. $\mathfrak{a} \subseteq (\mathfrak{a} : \mathfrak{b})$
2. $(\mathfrak{a} : \mathfrak{b})\mathfrak{b} \subseteq \mathfrak{a}$
3. $((\mathfrak{a} : \mathfrak{b}) : \mathfrak{c}) = (\mathfrak{a} : \mathfrak{b}\mathfrak{c}) = ((\mathfrak{a} : \mathfrak{c}) : \mathfrak{b})$
4. $(\bigcap_i \mathfrak{a}_i : \mathfrak{b}) = \bigcap_i (\mathfrak{a}_i : \mathfrak{b})$
5. $(\mathfrak{a} : \sum_i \mathfrak{b}_i) = \bigcap_i (\mathfrak{a} : \mathfrak{b}_i)$.

§2 Contraction and extension of ideal

Definition 2.1. *Let* R *be a ring and* $f : A \to B$ *be a ring homomorphism,*

1. *the* ***extension*** *of ideal* $\mathfrak{a}$ *of* A *is the ideal generated by* $f(\mathfrak{a})$ *in* B*, denoted by* $\mathfrak{a}^e$*.*
2. *the* ***contraction*** *of* $\mathfrak{b}$ *is* $f^{-1}(\mathfrak{b})$*, denoted by* $\mathfrak{b}^c$*.*

Especially if A *be a subring of* B *and* $i : A \to B$*, the contraction of ideal of* $\mathfrak{b}$ *of* B *is* $A \cap \mathfrak{b}$*.*

Proposition 2.2. .

1. $\mathfrak{a} \subseteq \mathfrak{a}^{ec}, \mathfrak{b} \supseteq \mathfrak{b}^{ce}$;
2. $\mathfrak{b}^c = \mathfrak{b}^{cec}, \mathfrak{a}^e = \mathfrak{a}^{ece}$;

3. *If $\mathcal{C}$ is the set of all contracted ideals in A and if $\mathcal{E}$ is the set of all extended ideals in B, then $\mathcal{C} = \{\mathfrak{a} \mid \mathfrak{a}^{ec} = \mathfrak{a}\}$, $\mathcal{E} = \{\mathfrak{b} \mid \mathfrak{b}^{ce} = \mathfrak{b}\}$, and $\mathfrak{a} \mapsto \mathfrak{a}^e$ is a bijective map, whose inverse is $\mathfrak{b} \mapsto \mathfrak{b}^c$.*

Proposition 2.3. .

$$\begin{array}{cc} (\mathfrak{a}_1 + \mathfrak{a}_2)^e = \mathfrak{a}_1^e + \mathfrak{a}_2^e & (\mathfrak{b}_1 + \mathfrak{b}_2)^c \supseteq \mathfrak{b}_1^c + \mathfrak{b}_2^c, \\ (\mathfrak{a}_1 \cap \mathfrak{a}_2)^e \subseteq \mathfrak{a}_1^e \cap \mathfrak{a}_2^e & (\mathfrak{b}_1 \cap \mathfrak{b}_2)^c = \mathfrak{b}_1^c \cap \mathfrak{b}_2^c, \\ (\mathfrak{a}_1 \mathfrak{a}_2)^e = \mathfrak{a}_1^e \mathfrak{a}_2^e & (\mathfrak{b}_1 \mathfrak{b}_2)^c \supseteq \mathfrak{b}_1^c \mathfrak{b}_2^c, \\ (\mathfrak{a}_1 : \mathfrak{a}_2)^e \subseteq (\mathfrak{a}_1^e : \mathfrak{a}_2^e) & (\mathfrak{b}_1 : \mathfrak{b}_2)^c \subseteq (\mathfrak{b}_1^c : \mathfrak{b}_2^c) \\ \mathrm{Rad}(\mathfrak{a})^e \subseteq \mathrm{Rad}\,(\mathfrak{a}^e) & \mathrm{Rad}(\mathfrak{b})^c = \mathrm{Rad}\,(\mathfrak{b}^c) \end{array}$$

if $\mathfrak{b}$ is a prime ideal in B, then so $\mathfrak{b}^c$.

The set $\mathcal{C}$ is closed under the other three operations, and $\mathcal{E}$ is closed under sum and product.

§3 Nil and nilpotent ideals

Definition 3.1. *Let A be a ring and $\mathfrak{a}$ be an (left, right, two-sided) ideal of A.*

1. *$\mathfrak{a}$ of A is **nil** if every element of $\mathfrak{a}$ is a nilpotent element;*

2. *$\mathfrak{a}$ is **nilpotent** if $\mathfrak{a}^n = 0$ for some integer n.*

Theorem 3.2. *Let A be a ring.*

1. *If $a \in A$ is nilpotent, a is both left and right quasiregula with quasi inverse $r = -a + a^2 - a^3 + \cdots + (-1)^{n-1} a^{n-1}$*

2. *Every nil left ideal is contained in $J(A)$.*

3. *Thus every nil ring is a radical ring.*

Proposition 3.3. *If R is a left [resp. right] Noetherian ring, then every nil ideal is nilpotent (Exercise 16).*

§4

Definition 4.1. *Let A be a ring and M be a left A-module.*

1. *The element $a \in A$ induces a homomorphism*

$$a_M : M \to M, \quad x \mapsto a \cdot x$$

   *called the **principal homomorphism associated with** a on M.*

2. *We say that a_M is **locally nilpotent** if for every $x \in M$, there exists a positive integer $n(x)$ such that $a^{n(x)} \cdot x = 0$.*

   Remark. *It is equivalent that*

Chapter II

Radical Ideals

Throughout this chapter, let A be a commutative ring.

Definition 0.1. *Let A be a commutative ring. If $\mathfrak{a}$ is any ideal of A, the ideal*

$$\mathrm{Rad}(\mathfrak{a}) := \{x \in A : x^n \in \mathfrak{a} \textit{ for some } n \in \mathbb{Z}_{\geq 1}\}$$

*is called **radical** of $\mathfrak{a}$, sometimes denoted by $\sqrt{\mathfrak{a}}$. The radical of 0 (the set of all nilpotent elements in A) is called **nilradical** of A, denoted by* $\mathrm{Nil}\,(A)$.

Proposition 0.2. *Then*

1. $\mathfrak{a} \subset r(\mathfrak{a})$
2. $r(r(\mathfrak{a})) = r(\mathfrak{a})$
3. $r(\mathfrak{a}\mathfrak{b}) = r(\mathfrak{a} \cap \mathfrak{b}) = r(\mathfrak{a}) \cap r(\mathfrak{b})$
4. *thus* $r\left(\mathfrak{a}_1\mathfrak{a}_2 \cdots \mathfrak{a}_n\right) = r\left(\bigcap \mathfrak{a}_i\right) = \bigcap r\left(\mathfrak{a}_i\right)$ *and* $r(\mathfrak{a}^n) = r(\mathfrak{a})$
5. $r(\mathfrak{a} + \mathfrak{b}) = r(r(\mathfrak{a}) + r(\mathfrak{b}))$
6. *if $\mathfrak{p}$ is prime in R, $r\left(\mathfrak{p}^n\right) = r\left(\mathfrak{p}\right) = \mathfrak{p}$ for all $n > 0$.*
7. $r(\mathfrak{a}) = (1) \Leftrightarrow \mathfrak{a} = (1)$

Proposition 0.3. *If S is a multiplicative subset which is disjoint from an ideal $\mathfrak{a}$, then there exists a ideal $\mathfrak{p}$ which is maximal in $\mathcal{S} = \{\mathfrak{b} : \mathfrak{a} \subset \mathfrak{b}$ and $\mathfrak{b} \cap S = \varnothing\}$. Furthermore, any such ideal $\mathfrak{p}$ is prime.*

Proof. Since $S \neq \varnothing$ and every ideal in $\mathcal{S}$ is properly contained in A, set $\mathcal{S}$ is partially ordered by inclusion. By Zorn's Lemma there is an ideal $\mathfrak{p}$ which is maximal in $\mathcal{S}$.

Let $\mathfrak{a}_1, \mathfrak{a}_2$ be ideals of A such that $\mathfrak{a}_1\mathfrak{a}_2 \subset \mathfrak{p}$. If $\mathfrak{a}_1 \not\subset \mathfrak{p}$ and $\mathfrak{a}_2 \not\subset \mathfrak{p}$, then each of the ideals $\mathfrak{p} + \mathfrak{a}_1$ and $\mathfrak{p} + \mathfrak{a}_2$ properly contains $\mathfrak{p}$ and hence must meet S. Consequently, for some $p_i \in \mathfrak{p}, a_i \in \mathfrak{a}_i$.

$$p_1 + a_1 = s_1 \in S \quad \text{and} \quad p_2 + a_2 = s_2 \in S$$

Thus $s_1 s_2 = p_1 p_2 + p_1 a_2 + a_1 p_2 + a_1 a_2 \in \mathfrak{p} + \mathfrak{a}_1 \mathfrak{a}_2 \subset \mathfrak{p}$. This is a contradiction since $s_1 s_2 \in S$ and $S \cap \mathfrak{p} = \varnothing$. Therefore $\mathfrak{a}_1 \subset \mathfrak{p}$ or $\mathfrak{a}_2 \subset \mathfrak{p}$, whence $\mathfrak{p}$ is prime. $\square$

Theorem 0.4. *Let A be a commutative rng and an ideal $\mathfrak{a}$.*

1. *If $\pi : A \to A/\mathfrak{a}$ is the canonical projection, then* $\operatorname{Rad}(\mathfrak{a}) = \pi^{-1}(\operatorname{Nil}(A/\mathfrak{a}))$
2. *The radical of an ideal $\mathfrak{a}$ is the intersection of the prime ideals which contain $\mathfrak{a}$, that is,*

$$\operatorname{Rad}(\mathfrak{a}) = \bigcap_{\substack{\mathfrak{a} \subset \mathfrak{p}, \\ \mathfrak{p} \in \operatorname{Spec} A}} \mathfrak{p}$$

Proof. It is clear that

$$\operatorname{Rad}(\mathfrak{a}) \subset \bigcap_{\substack{\mathfrak{a} \subset \mathfrak{p} \\ \mathfrak{p} \text{ is prime}}} \mathfrak{p} := \tilde{\mathfrak{p}}$$

If $S = \tilde{\mathfrak{p}} - \operatorname{Rad}(\mathfrak{a})$ is nonempty, whence is a multiplicative subset of R (verify that $x, y \in S \Rightarrow xy \in S$) and disjoint from $\operatorname{Rad}(\mathfrak{a})$, there exist a prime ideal $\mathfrak{p}'$ that contains $\operatorname{Rad}(\mathfrak{a})$ and disjoint from S by 0.3. But $\tilde{\mathfrak{p}} \subset \mathfrak{p}'$ by the definition of $\tilde{\mathfrak{p}}$, this is a contradiction. $\square$

Proposition 0.5. *If A is a commutative ring with identity $\neq 0$, then $A^\times + \operatorname{Nil}(A) \subset A^\times$.*

Chapter III

Fractions and Localization

§1 Rings of Quotient

A is a commutative ring with identity throughout this section unless otherwise stated.

Definition 1.1. *Let A be a commutative ring. A subset S called a **multiplicative subset** of A if S is a submonoid of $(A, \times)$.*

Remark. *In general, we always assume that $0 \notin S$.*

Definition 1.2. *Let S be a multiplicative subset of A and*

1. *The relation defined on the set $A \times S$ by*

$$(a, s) \sim (a', s') \Leftrightarrow s_1 (as' - a's) = 0 \text{ for some } s_1 \in S$$

   is an equivalence relation and the equivalence class containing the element (a, s) is denoted by a/s.

2. *$S^{-1}R$ is a commutative ring with identity $1/1$, where addition and multiplication are defined by*

$$r/s + r'/s' = (rs' + r's)/ss' \quad \text{and} \quad (r/s)(r'/s') = rr'/ss'$$

   *is called the **ring of fractions** of R by S.*

Remark. *The map $\varphi_S : A \to S^{-1}A$ given by $a \mapsto a/1$ is a well-defined homomorphism of rings and $\varphi(S) \subset (S^{-1}A)^{\times}$.*

Theorem 1.3 (Universal property)**.** *Let $\mathcal{C}$ be the category*

- *whose objects are ring-homomorphisms (commutative rings with identity)*

$$f : A \to B$$

  such that for every $s \in S$, the element $f(s)$ is invertible in B.

- *If $f : A \to B$ and $f' : A \to B'$ are two objects of $\mathcal{C}$, a morphism g of f into f' is a homomorphism*
$$g : B \to B'$$
making the diagram commutative:

We have that $\varphi_S : A \to S^{-1}A$ is a universal object in this category $\mathcal{C}$.

Theorem 1.4. *Let S be a multiplicative subset of A.*

1. *If S contains no zero divisors, then φ_S is a monomorphism.*
2. *If A has no zero divisors and $0 \notin S$, then $S^{-1}A$ is an integral domain.*
3. *If $S \subset A^\times$, then φ_S is an isomorphism.*

Definition 1.5. *Let A be a commutative ring and S be the set of all nonzero elements of A that are not zero divisors, then $S^{-1}A$ is called the* ***complete ring of quotients*** *of the ring A.*

The complete ring of quotients of an integral domain A is its ***quotient field****, denoted by* $\operatorname{Frac}(A)$.

§2 Extensions and Contractions in ring of fractions

Let A be a commutative with identity 1_A and S be a multiplicative subset with $\varphi_S : A \to S^{-1}A$

Proposition 2.1.

1. *If $\mathfrak{a}$ is an ideal in A, then $S^{-1}\mathfrak{a} = \{a/s \mid a \in \mathfrak{a}; s \in S\} = \mathfrak{a}S^{-1}A = \mathfrak{a}^e$.*
2. *If $\mathfrak{b}$ is an ideal in $S^{-1}A$, then $\varphi_S^{-1}(\mathfrak{b})$ coincides with $\mathfrak{b}^c$*
3. *let $\mathfrak{a}$ be an ideal of A, then $S^{-1}\mathfrak{a} = S^{-1}A$ if and only if $S \cap \mathfrak{a} \neq \varnothing$.*

Corollary 2.2.
$$S^{-1}(\mathfrak{a}+\mathfrak{b}) = S^{-1}\mathfrak{a} + S^{-1}\mathfrak{b}$$
$$S^{-1}(\mathfrak{a}\mathfrak{b}) = \left(S^{-1}\mathfrak{a}\right)\left(S^{-1}\mathfrak{b}\right)$$
$$S^{-1}(\mathfrak{a}\cap\mathfrak{b}) = S^{-1}\mathfrak{a} \cap S^{-1}\mathfrak{b}$$
$$S^{-1}\operatorname{Rad}(\mathfrak{a}) = \operatorname{Rad}\left(S^{-1}\mathfrak{a}\right)$$

Theorem 2.3. .

1. *$\mathfrak{b}^{ce} = \mathfrak{b}$ for all ideals $\mathfrak{b}$ of $S^{-1}A$. In other words every ideal in $S^{-1}A$ is of the $S^{-1}\mathfrak{a} = \mathfrak{a}^e$ for some ideal $\mathfrak{a}$ in A by proposition 2.2 .*
2. *$\mathfrak{a}^{ec} = \bigcup_{s\in S}(\mathfrak{a} : s)$*
3. *If $\mathfrak{p}$ is a prime ideal in A and $S \cap \mathfrak{p} = \varnothing$, then $S^{-1}\mathfrak{p}$ is a prime ideal in $S^{-1}A$*
4. *there is a one-to-one correspondence between the set $\mathcal{U} = \{\mathfrak{p} : \mathfrak{p}$ is prime and disjoint from $S\}$ and the set $\mathcal{V} = \{S^{-1}\mathfrak{p} : S^{-1}\mathfrak{p}$ is prime in $S^{-1}R\}$, given by $\mathfrak{p} \mapsto S^{-1}\mathfrak{p}$.*

Proof. Let $I = \varphi_S^{-1}(J)$, then $I^e = J^{ce} \subset J$, whence $S^{-1}I \subset J$. Conversely, if $r/s \in J$, then $\varphi_S(r) = rs/s = (r/s)\,(s^2/s) \in J$, whence $r \in \varphi_S^{-1}(J) = I$. Thus $r/s \in S^{-1}I$ and hence $J \subset S^{-1}I$. □

§3 Fractions of modules

Let A be a commutative ring with identity, M be a A-module and S be a multiplicative subset of A.

Definition 3.1. *The* ***module of fractions*** *of M with respect to S is the set*

$$S^{-1}M = \{m/s \mid m \in M, s \in S\}$$

of equivalence classes of the relation on $M \times S$ defined by

$$(m,s) \sim (m',s') \Leftrightarrow s_1\,(s'm - sm') = 0 \textit{ for some } s_1 \in S$$

with addition and scalar multiplication defined by

$$m/s + m'/s' = (s'm + sm')\,/ss' \quad \textit{and} \quad (a/t)(m/s) = (am)/ts$$

for all $m, m' \in M$, $s, s', t \in S$ and $a \in A$.

Lemma 3.2. *1.* $S^{-1}(N+P) = S^{-1}(N) + S^{-1}(P)$

2. $S^{-1}(N \cap P) = S^{-1}(N) \cap S^{-1}(P)$

Proposition 3.3. .

1. *The functor $S^{-1} : {}_A\mathbf{Mod} \to {}_{S^{-1}A}\mathbf{Mod}$ is exact*
2. *$S^{-1}\square$ is additive*
3. *$S^{-1}\square \cong S^{-1}A \otimes_A \square$ in $[{}_A\mathbf{Mod}, {}_{S^{-1}A}\mathbf{Mod}]$*

Corollary 3.4. .

1. *$S^{-1}(M/N) \cong (S^{-1}M)\,/\,(S^{-1}N)$ in ${}_{S^{-1}A}\mathbf{Mod}$*
2. *$S^{-1}M \otimes_{S^{-1}A} S^{-1}N \cong S^{-1}(M \oplus_A N)$ in ${}_{S^{-1}A}\mathbf{Mod}$*
3. *$S^{-1}A$ is a flat A-module*

§4 Localization and Local rings

Definition 4.1. *Let A be a commutative ring with identity, $\mathfrak{p}$ a prime ideal of A and multiplicative subset $S = A - \mathfrak{p}$. The ring of quotients $S^{-1}A$ is called the **localization of** A **at** $\mathfrak{p}$ and is denoted $A_\mathfrak{p}$. If $\mathfrak{a}$ is an ideal in A, then the ideal $\mathfrak{a}^e = S^{-1}\mathfrak{a}$ in $A_\mathfrak{p}$.*

Remark. *We always identify A with its image $\varphi_S(A)$ in $A_\mathfrak{p}$ thus A can be considered as a subring of $A_\mathfrak{p}$. In this case, the extension ideal $\mathfrak{a}^e = S^{-1}\mathfrak{a} = \mathfrak{a}A_\mathfrak{p}$.*

Theorem 4.2. *Let $\mathfrak{p}$ be a prime ideal in a commutative ring A with identity and localization $A_\mathfrak{p}$.*

1. *There is a one-to-one correspondence between the set $\{\mathfrak{q} : \mathfrak{q}$ is prime and contained in $\mathfrak{p}\}$ and the set $\{S^{-1}\mathfrak{q} : S^{-1}\mathfrak{q}$ is prime in $A_\mathfrak{p}\}$, given by $\mathfrak{q} \mapsto S^{-1}\mathfrak{q}$;*
2. *The ideal $S^{-1}\mathfrak{p}$ is the unique maximal ideal of $A_\mathfrak{p}$.*

Definition 4.3. *A **local ring** is a commutative ring with identity which has a unique maximal ideal.*

Theorem 4.4. *The following conditions are equivalent.*

1. *R is a local ring;*
2. *all nonunits of R are contained in some ideal $M \neq R$;*
3. *the nonunits of R form an ideal.*
4. *for all $r, s \in R$, $r + s = 1_R$ implies r or s is a unit.*

Proposition 4.5. *Every nonzero homomorphic image of a local ring is local.*

§5 Local properties

Definition 5.1. *A property P of a ring A (or of an A-module M) is said to be a **local property** if the following is true: A (or M) has $P \Leftrightarrow A_\mathfrak{p}$ (or $M_\mathfrak{p}$) has P, for each prime ideal $\mathfrak{p}$ of A.*

Proposition 5.2. *The following are local properties:*

1. *A-module M is zero*
2. *A-module homomorphism $\phi : M \to N$ is injective [resp. surjective, bijective]*
3. *A-module M is flat*

Chapter IV

Chain Condition

§1

Definition 1.1. *Let A be a ring.*

1. *A left A-module M is said to satisfy the* ***ascending chain condition (ACC) on submodules*** *(or to be* ***Noetherian****) if for every chain $M_1 \subset M_2 \subset M_3 \subset \cdots$ of submodules of M, there is an integer n such that $M_i = M_n$ for all $i \geq n$.*
2. *The ring A is* ***left Noetherian*** *if left regular module ${}_AA$ is Noetherian. A is said to be* ***Noetherian*** *if A is both left and right Noetherian.*
3. *A left A-module N is said to satisfy, the* ***descending chain condition (DCC) on submodules*** *(or to be* ***Artinian****) if for every chain $N_1 \supset N_2 \supset N_3 \supset \cdots$ of submodules of N, there is an integer m such that $N_i = N_m$ for all $i \geq m$.*
4. *A is* ***left Artinian*** *if ${}_AA$ is Artinian.*

   A is said to be ***Artinian*** *if A is both left and right Artinian.*

Definition 1.2. *Let A be a ring, a left A-module M is said to satisfy the* ***maximum condition*** *[resp. minimum condition] on submodules if every nonempty set of submodules of M contains a maximal [resp. minimal] element (with respect to set theoretic inclusion).*

Theorem 1.3. *A module satisfies the ascending [resp. descending] chain condition on submodules if and only if A satisfies the maximal [resp. minimal] condition on submodules.*

Theorem 1.4. *Let A be a ring and $0 \to M' \to M \to M'' \to 0$ be a short exact sequence of left A-modules. Then ${}_AM$ is Noetherian [resp. Artinian] if and only if ${}_AM'$ and ${}_AM''$ are both Noetherian [resp. Artinian].*

Corollary 1.5 (Stability of chain conditions)**.** *Let A be a ring, we have*

1. *Let M be an A-module and $M_1 \subseteq M$ a submodule. Then M is Noetherian [resp. Artinian] if and only if both M_1 and M/M_1 are Noetherian [resp. Artinian].*

2. *For A-modules $M_1, \ldots, M_n$, the direct sum*

$$M_1 \oplus \cdots \oplus M_n$$

is Noetherian [resp. Artinian] if and only if each M_i is.

3. *Any homomorphic image of a Noetherian [resp. Artinian] module is Noetherian [resp. Artinian].*

4. *If A is a left Noetherian [resp. Artinian] ring, then every finitely generated left A-module is Noetherian [resp. Artinian].*

§2 Normal series and Composition Series of Modules

Definition 2.1. *Let A be a ring and a left A-module M.*

1. *A **normal series** for M is a chain of submodules: $M = M_0 \supset M_1 \supset M_2 \supset \cdots \supset M_n$. The factors of the series are the quotient modules*

$$M_i/M_{i+1} \quad (i = 0, 1, \ldots, n-1).$$

*The **length** of the normal series is the number of proper inclusions (= number of nontrivial factors).*

2. *A **refinement** of the normal series $M = M_0 \supset M_1 \supset \cdots \supset M_n$ is a normal series obtained by inserting a finite number of additional submodules between the given ones. A **proper refinement** is one which has length larger than the original series.*

3. *Two normal series are **equivalent** if there is a one-to-one correspondence between the nontrivial factors such that corresponding factors are isomorphic modules. Thus equivalent series necessarily have the same length.*

4. *A **composition series** for A is a normal series $A = A_0 \supset A_1 \supset A_2 \supset \cdots \supset A_n = 0$ such that each factor A_k/A_{k+1} $(k = 0, 1, \ldots, n-1)$ is a module with no proper nonempty submodules.*

Theorem 2.2. *Any two normal series of a module A have refinements that are equivalent. Any two composition series of A are equivalent.*

Theorem 2.3. *Let A be a ring. A left A-module M has a composition series if and only if M satisfies both the ACC and DCC on submodules.*

Proof. $(\Rightarrow)$ Suppose A has a composition series S of length n. If either chain condition fails to hold, one, can find submodules

$$A = A_0 \supsetneq A_1 \supsetneq A_2 \supsetneq \cdots \supsetneq A_n \supsetneq A_{n+1}$$

which form a normal series T of length $n+1$. By 2.2, S and T have refinements that are equivalent. This is a contradiction since equivalent series have equal length. For every refinement of the composition series S has the same length n as S, but every refinement of T necessarily has length at least $n+1$. Therefore, A satisfies both chain conditions.

$(\Leftarrow)$ If B is a nonzero submodule of A, let $S(B)$ be the set of all proper submodules C of B. Also define $S(0)=\{0\}$. For each B there is a maximal element B' of $S(B)$ by 1.3. Let S be the set of all submodules of A and define a map $f: S \to S$ by $f(B)=B'$

Let $A_i = f^{(i)}(A)$, then $A \supset A_1 \supset A_2 \supset \cdots$ is a descending chain by construction, whence for some n, $A_i = A_n$ for all $i \geq n$. Since $A_{n+1} = f(A_n)$, the definition of f shows that $A_{n+1} = A_n$ only if $A_n = 0 = A_{n+1}$. Let m be the smallest integer such that $A_m = 0$. Then $m \leq n$ and $A_k \neq 0$ for all $k < m$. Furthermore for each $k < m$, A_{k+1} is a maximal submodule of A_k such that $A_k \supsetneq A_{k+1}$. Consequently, each A_k/A_{k+1} is nonzero and has no proper submodules by **??**. Therefore, $A \supset A_1 \supset \cdots \supset A_m = 0$ is a composition series for A. □

Chapter V

Noetherian Ringa and Modules

§1 Basic Criterion

§1.1 General case

theorem 1.3 corollary 1.5

Proposition 1.1. *A left A-module M is Noetherian if and only if every submodule of M is finitely generated.*

§1.2 Commutative case

Proposition 1.2. *Let A be a commutative Noetherian and let S be a multiplicative subset of A. Then the ring of fractions $S^{-1}A$ is Noetherian.*

Theorem 1.3. *Let A be a left Noetherian ring, then $A[x_1, \ldots, x_n]$ is also left Noetherian.*

Proof. We prove the case $n = 1$; the general case follows by induction on n.

Let $\mathfrak{a}$ be an left ideal of $A[x]$. For each nonnegative integer d, let $\mathfrak{a}_d$ be the set of all leading coefficients of polynomials in $\mathfrak{a}$ of degree d together with 0. Then $\mathfrak{a}_d$ is a left ideal of A for each d and $\mathfrak{a}_0 \subset \mathfrak{a}_1 \subset \mathfrak{a}_2 \subset \cdots$ is an ascending chain of left ideals of A. Since A is left Noetherian, there exists an integer n such that $\mathfrak{a}_d = \mathfrak{a}_n$ for all $d \geq n$.

For each $d = 0, 1, \ldots, n$, let $a_{d1}, a_{d2}, \ldots, a_{dm_d}$ be generators of the left ideal $\mathfrak{a}_d$. For each a_{di}, choose a polynomial $f_{di}(x) \in \mathfrak{a}$ of degree d with leading coefficient a_{di}.

We claim that the finite set of polynomials

$$\{f_{di}(x) \mid d = 0, 1, \ldots, n; i = 1, 2, \ldots, m_d\}$$

generates the ideal $\mathfrak{a}$.

Let $f(x)$ be any polynomial in $\mathfrak{a}$. If deg $f(x) = d$, then the leading coefficient of $f(x)$ is in

$\mathfrak{a}_d = \mathfrak{a}_n$, so there exist elements $b_i \in A$ such that

$$\text{leading coefficient of } f(x) = \sum_{i=1}^{m_n} b_i a_{ni}$$

Let $g(x) = \sum_{i=1}^{m_n} b_i x^{d-n} f_{ni}(x)$. Then $g(x) \in \mathfrak{a}$ and $\deg g(x) = d$ with leading coefficient equal to that of $f(x)$. Thus $\deg(f(x) - g(x)) < d$ and by repeating this process, we can express $f(x)$ as an $A[x]$-linear combination of the polynomials $f_{di}(x)$. Therefore, $\mathfrak{a}$ is finitely generated and $A[x]$ is left Noetherian. □

Corollary 1.4. *The finitely generated algebra over a left Noetherian ring is left Noetherian.*

Proposition 1.5. *Let $A \subseteq B \subseteq C$ be rings. Suppose that A is Noetherian, that C is finitely generated as an A-algebra and that C is either (i) finitely generated as a B-module or (ii) integral over B. Then B is finitely generated as an A-algebra.*

Proposition 1.6. *Let A be a subring of B; suppose that A is Noetherian and that B is finitely generated as an A-module. Then B is a Noetherian ring.*

Proposition 1.7. *If A is Noetherian and S is any multiplicatively closed subset of A, then $S^{-1}A$ is Noetherian.*

Theorem 1.8 (Hilbert's Basis Theorem)**.** *If A is a commutative Noetherian ring with identity, then so is $R[x_1, \ldots, x_n]$ and $R[[x]]$*

Proposition 1.9. *If R is a commutative ring with identity and $\mathfrak{p}$ is an ideal which is maximal in the set of all ideals of R which are not finitely generated, then $\mathfrak{p}$ is prime.*

Proof. Suppose $ab \in \mathfrak{p}$ but $a \notin \mathfrak{p}$ and $b \notin \mathfrak{p}$. Then $\mathfrak{p} + (a)$ and $\mathfrak{p} + (b)$ are ideals properly containing $\mathfrak{p}$ and therefore finitely generated by maximality of $\mathfrak{p}$. Consequently $\mathfrak{p} + (a) = (p_1 + r_1 a, \ldots, p_n + r_n a)$ and $\mathfrak{p} + (b) = (p_1' + r_1' b, \ldots p_m' + r_m' b)$ for some $p_i, p_i' \in \mathfrak{p}$ and $r_i, r_i' \in R$.

If $J = (\mathfrak{p} : a) = \{r \in R \mid ra \in \mathfrak{p}\}$, then J is an ideal. Since $ab \in \mathfrak{p}$, $(p_i{}' + r_i{}' b)\, a = p_i{}' a + r_i{}' ab \in \mathfrak{p}$ for all i, whence $\mathfrak{p} \subset \mathfrak{p} + (b) \subset J$. By maximality, J is finitely generated, say $J = (j_1, \ldots, j_k)$.

If $x \in \mathfrak{p}$, then $x \in \mathfrak{p} + (a)$ and hence for some $s_i \in R$, $x = \sum_{i=1}^{n} s_i (p_i + r_i a) = \sum_{i=1}^{n} s_i p_i + \sum_{i=1}^{n} s_i r_i a$. Consequently, $(\sum_i s_i r_i)\, a = x - \sum_i s_i p_i \in \mathfrak{p}$, whence $\sum_i s_i r_i \in J$. Thus for some $t_i \in R$, $\sum_{i=1}^{n} s_i r_i = \sum_{i=1}^{k} t_i j_i$ and $x = \sum_{i=1}^{n} s_i p_i + \sum_{i=1}^{k} t_i j_i a$. Therefore, $\mathfrak{p}$ is generated by $p_1, \ldots, p_n, j_1 a, \ldots, j_k a$, which is a contradiction. Thus $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$ and $\mathfrak{p}$ is prime by **??** □

Theorem 1.10 (I.S.Cohen)**.** *A commutative ring R with identity is Noetherian if and only if every prime ideal of R is finitely generated.*

Proof. Let $\mathcal{S}$ be the set of all ideals of R which are not finitely generated. If $\mathcal{S}$ is nonempty, then use Zorn's Lemma to find a maximal element P of $S.P$ is prime by Proposition 2.4 and hence finitely generated by hypothesis.

This is a contradiction unless $\delta = \varnothing$. Therefore, R is Noetherian by Theorem 1.9. □

§2 Primary Decomposition

Throught this section, A be a commutative ring, and that module [resp. homomorphism] are A-modules [resp. A-module homomorphisms] unless otherwise stated.

Definition 2.1. *Let M be an A-module.*

1. *A submodule Q of M is said to be* ***primary*** *if $Q \neq M$, and if principal homomorphism $a_{M/Q}$ is injective or nilpotent for each $a \in A$.*

   A ideal $\mathfrak{q}$ in A is said to be ***primary*** *if it is a primary submodule of the regular module ${}_A A$.*

   Remark. *Ideal $\mathfrak{q}$ is primary in A iff $xy \in \mathfrak{q}, x \notin \mathfrak{q} \Rightarrow y^n \in \mathfrak{q}$ for some $n > 0$*

2. *Let Q be a primary submodule of M. The* ***associated prime ideal*** *of Q is the prime ideal $\mathfrak{p} = \operatorname{Rad}(\operatorname{Ann}(M/Q))$. We shall say that Q is* ***primary for*** *$\mathfrak{p}$ or that Q is $\mathfrak{p}$-**primary**.*

Proposition 2.2. *If $Q_1, \ldots, q_n$ are $\mathfrak{p}$-primary submodules of M, then so is their intersection $Q_1 \cap Q_2 \cap \cdots \cap Q_n$.*

Theorem 2.3. *Let A be a commutative ring, $\mathfrak{q}$ and $\mathfrak{p}$ be ideals in A. Then $\mathfrak{q}$ is primary for $\mathfrak{p}$ if and only if*

(i) $\mathfrak{q} \subset \mathfrak{p} \subset \operatorname{Rad}(\mathfrak{q})$

(ii) if $ab \in \mathfrak{q}$ and $a \notin \mathfrak{q}$, then $b \in \mathfrak{p}$.

Proof. Suppose (i) and (ii) hold. If $ab \in \mathfrak{q}$ with $a \notin \mathfrak{q}$, then $b \in \mathfrak{p} \subset \operatorname{Rad}\mathfrak{q}$, whence $b^n \in \mathfrak{q}$ for some $n > 0$. Therefore $\mathfrak{q}$ is primary.

To show that $\mathfrak{q}$ is primary for $\mathfrak{p}$ we need only show $\mathfrak{p} = \operatorname{Rad}\mathfrak{q}$. By (i), $\mathfrak{p} \subset \operatorname{Rad}\mathfrak{q}$. If $b \in \operatorname{Rad}\mathfrak{q}$, let n be the least integer such that $b^n \in \mathfrak{q}$. If $n = 1, b \in \mathfrak{q} \subset \mathfrak{p}$. If $n > 1$, then $b^{n-1}b = b^n \in \mathfrak{q}$, with $b^{n-1} \notin \mathfrak{q}$ by the minimality of n. By (ii), $b \in \mathfrak{p}$. Thus $b \in \operatorname{Rad}\mathfrak{q}$ implies $b \in \mathfrak{p}$, whence $\operatorname{Rad}\mathfrak{q} \subset \mathfrak{p}$. □

Proposition 2.4. *Clearly every prime ideal is primary. Also the contraction of a primary ideal is primary, for if $f : A \to B$ and if $\mathfrak{q}$ is a primary ideal in B, then $A/\mathfrak{q}^c$ is isomorphic to a subring of B/q.*

Proposition 2.5. *If $\operatorname{Rad}(\mathfrak{a})$ is maximal, then $\mathfrak{a}$ is primary. In particular, the powers of a maximal ideal $\mathfrak{m}$ are $\mathfrak{m}$-primary.*

Proof. Let $\operatorname{Rad}(\mathfrak{a}) = \mathfrak{m}$. The image of $\mathfrak{m}$ in $A/\mathfrak{a}$ is the nilradical of $A/\mathfrak{a}$, hence $A/\mathfrak{a}$ has only one prime ideal $\pi(\mathfrak{m})$, by (1.8). Hence every element of $A/\mathfrak{a}$ is either a unit or nilpotent, and so every zero-divisor in A/a is nilpotent. □

Definition 2.6. *Let N be a submodule of an A-module M.*

1. *When N is written as a finite intersection of primary submodules of M,*

$$N = Q_1 \cap Q_2 \cap \cdots \cap Q_n$$

*we shall say that N has a **primary decomposition** in M.*

2. *If each associated prime ideal $\mathfrak{p}_i$ of Q_i is distinct, then the primary decomposition is said to be **reduced**.*

3. *If $\mathfrak{p}_i$ does not contain any $\mathfrak{p}_j$ for $j \neq i$, then $\mathfrak{p}_i$ is called an **isolated prime** of the primary decomposition.*

Remark. *Using 2.3, we can always obtain from a given primary decomposition of N another reduced primary decomposition.*

Theorem 2.7 (Uniqueness of reduced primary decomposition)**.** *Let N be a submodule of M with two reduced primary decompositions,*

$$Q_1 \cap Q_2 \cap \cdots \cap Q_k = N = Q_1' \cap Q_2' \cap \cdots \cap Q_s'$$

where Q_i is $\mathfrak{p}_i$-primary and Q_j' is $\mathfrak{p}_j'$-primary. Then $k = s$ and (after reordering if necessary) $\mathfrak{p}_i = \mathfrak{p}_i'$ for all $i = 1, 2, \ldots, k$. Furthermore if Q_i and Q_i' both are $\mathfrak{p}_i$-primary and $\mathfrak{p}_i$ is an isolated prime, then $Q_i = Q_i'$.

Theorem 2.8 (Existence of reduced primary decomposition)**.** *Let M be a Noetherian A-module and N a submodule of M. Then N admits a reduced primary decomposition.*

Proof. Let $\mathcal{S}$ be the set of all submodules of M that do not have a primary decomposition. Clearly no primary submodule is in $\mathcal{S}$. We must show that $\mathcal{S}$ is actually empty. If $\mathcal{S}$ is nonempty, then $\mathcal{S}$ contains a maximal element C by Theorem 1.4.

Since C is not primary, there exist $r \in R$ and $b \in M - C$ such that $rb \in C$ but $r^n M \not\subset C$ for all $n > 0$. Let $M_n = (C : r^n) = \{x \in M \mid r^n x \in C\}$. Then each M_n is a submodule of M and $M_1 \subset M_2 \subset M_3 \subset \cdots$. By hypothesis there exists $k > 0$ such that $M_i = M_k$ for $i \geq k$. Let D be the submodule $r^k M + C = \left\{x \in M \mid x = r^k y + c \text{ for some } y \in M, c \in C\right\}$. Clearly $C \subset M_k \cap D$.

Conversely, if $x \in M_k \cap D$, then $x = r^k y + c$ and $r^k x \in C$, whence $r^{2k} v = r^k \left(r^k y\right) = r^k (x - c) = r^k x - r^k c \in C$. Therefore, $y \in M_{2k} = M_k$. Consequently, $r^k y \in C$ and hence $x = r^k y + c \in C$. Therefore $M_k \cap D \subset C$, whence $M_k \cap D = C$. Now $C \neq M_k \neq M$ and $C \neq D \neq M$ since $b \in M_k - C$ and $r^k M \not\subset C$. By the maximality of C in $\mathcal{S}$, M_k and D must have primary decompositions. Thus C has a primary decomposition, which is a contradiction. Therefore $\mathcal{S}$ is empty and every submodule has a primary decomposition. Consequently, every submodule has a reduced primary decomposition by **??**. □

Corollary 2.9. *If R is a commutative Noetherian ring with identity and M is a finitely generated unitary R-module. Then every submodule $N (\neq M)$ of M has a reduced primary decomposition.*

Proof. This is an immediate consequence of **?? ??** and 2.8 □

Chapter VI

Artinian

§1 Artinian Rings

Lemma 1.1. *If A is a left Artinian ring, then $J(A)$ is nilpotent.*

Proof. Consider the descending chain of ideals

$$J \supset J^2 \supset J^3 \supset \cdots$$

there exists an integer n such that $J^n = J^{n+1} = \cdots$. Denote J^n by I, then $I = JI$. Assume $I \neq 0$ and let nonempty set $I^2 = I \neq 0$

$$\mathcal{S} := \{\text{left ideal } L : IL \neq 0\}$$

Then there exists a minimal element L_0 in $\mathcal{S}$ by 1.3. Furthermore, $L_0 = Aa$ for some $a \in L_0$ and $IL_0 = L_0$ by minimality. Hence, there exists a nonzero $x \in I \subset J^n$ such that $xa = a$, that is, $1_A - x$ is a left zero divisor which contradicts the definition of J. Therefore, $J^n = I = 0$. □

Lemma 1.2. *Let A is simple left Artinian ring, then the following condition are equivalent for a left A-module M:*

1. *Noetherian*
2. *Artinian*
3. *has finite length.*

Theorem 1.3 (Hopkins-Levitzki)**.** *Let A be a left Artinian ring and a left A-modules M. If M is Artinian, then M is Noetherian.*

Proof. Consider the descending chain of submodules

$$M \supset JM \supset J^2M \supset \cdots \supset J^nM = 0$$

where $J = J(A)$. Then $J^k M$ and $J^k M/J^{k+1}M$ are Artinian for all k by 1.5.

On the other hand, $J^k M/J^{k+1}M$ is a module over the semisimple Artinian ring A/J, and is also Artinian. Thus each $J^k M/J^{k+1}M$ is Noetherian and thus M is Noetherian by 1.5. □

Example 1.4. *Let ring* $A = \begin{pmatrix} \mathbb{Q} & \mathbb{R} \\ 0 & \mathbb{R} \end{pmatrix}$*, A is right Artinian but not left Artinian. infinite descending chain of left ideals*

$$\begin{pmatrix} 0 & V_n \\ 0 & 0 \end{pmatrix} \supsetneq \begin{pmatrix} 0 & V_{n+1} \\ 0 & 0 \end{pmatrix} \supsetneq \cdots$$

where $\{V_n\}$ is a infinite descending chain of $\mathbb{Q}$-subspace of $\mathbb{R}$.

In generally, the ring of upper triangular matrices

$$T := \begin{pmatrix} D_1 & M \\ 0 & D_2 \end{pmatrix}$$

where D_1, D_2 are rings and M is a (D_1, D_2)-bimodule. Then T is left Artinian if and only if D_1, D_2 are left Artinian and M_{D_2} has finite length.

§2 Commutative Artinian Rings

In this subsection we shall study commutative Artinian rings.

Theorem 2.1. *Let A be a commutative ring with identity. Then the following conditions are equivalent:*

1. *A is Artinian*
2. *A is Noetherian and* $\dim A = 0$.
3. $\operatorname{Spec}(A)$ *is finite and Krull* $\dim A = 0$
4. *(Structure theorem) A is isomorphic to a finite direct product of Artin local rings.*

Proof. 1. ⇒ 2. By 1.3, A is Noetherian. If $\mathfrak{p}$ is a prime ideal in A, then $A/\mathfrak{p}$ is an integral domain and Artinian, whence a field. Thus $\dim A = 0$.

2. ⇒ 3. Let $\mathcal{S} := \{\mathfrak{p}_1 \cap \mathfrak{p}_1 \cdots \cap \mathfrak{p}_r : \mathfrak{p}_i \text{ maximal ideal}\}$, then $\mathcal{S}$ has a minimal element $\mathfrak{a} = \mathfrak{p}_1 \cap \mathfrak{p}_1 \cdots \cap \mathfrak{p}_n$ by 1.3. Hence for any maximal ideal $\mathfrak{p}$, $\mathfrak{p} \supset \mathfrak{a}$ by minimality of $\mathfrak{a}$, whence $\mathfrak{m} = \mathfrak{m}_i$ for some i by . □

Corollary 2.2. *In a commutative Artin ring the*

1. *nilradical is equal to the Jacobson radical.*
2. *A has finitely many maximal ideals*

Proposition 2.3. *Let A be an Artin local ring. Then the following are equivalent:*

1. *every ideal in A is principal;*

2. *the maximal ideal $\mathfrak{m}$ is principal;*

3. $\dim_k(\mathfrak{m}/\mathfrak{m}^2) \leqslant 1$.

Chapter VII

The Structure of Rings and Modules

§1 Simplicity

Definition 1.1. *A ring A is said to be **simple** if A has no proper two-sided ideals.*

Definition 1.2. *A left module M over a ring A is said to be **simple** (or **irreducible**) if M has no proper submodules.*

Remark. *A left ideal $\mathfrak{a}$ of A is a simple left A-module if and only if $\mathfrak{a}$ is a minimal left ideal of A. In this case, we call $\mathfrak{a}$ the **simple left ideal** of A.*

Proposition 1.3. *Let A be a ring and M be a simple left A-module, then*

1. *$M = Am$ for every $0 \neq m \in M$.*
2. *If $0 \neq u \in M$, then $M \cong A/\operatorname{Ann}(u)$, thus $\operatorname{Ann}(u)$ is a left maximal ideal.*

   Conversely, if $\mathfrak{m}$ is left maximal in A, then $A/\mathfrak{m}$ is a simple A-module with $\operatorname{Ann}(A/\mathfrak{m}) = \mathfrak{m}$
3. *If A is not a division ring, then M is a torsion module.$m \in M$.*

Lemma 1.4 (Schur)**.** *Let M be a simple left module over a ring A and let N_i be any A-module.*

1. *Every nonzero A-module homomorphism $f : M \to N_1$ is a monomorphism;*
2. *Every nonzero A-module homomorphism $g : N_2 \to M$ is an epimorphism;*
3. *The endomorphism ring $D = \operatorname{Hom}_A(M, M)$ is a division ring, then M is a vector space over $\operatorname{Hom}_A(M, M)$ with $fa = f(a)$*

§1.1 Primitivity

Definition 1.5. *Let A be a ring.*

1. *The ring A is said to be **left primitive** if there exists a simple faithful left A-module.*

2. *An two-sided ideal $\mathfrak{a}$ of a ring A is said to be* ***left primitive*** *if the quotient ring $A/\mathfrak{a}$ is a left primitive ring.*

Remark. *If M is a simple left A-module, then $A/\operatorname{Ann}(M)$ is left primitive with faithful simple left $A/\operatorname{Ann}(M)$-module M.*

Proposition 1.6. *Let A be a ring.*

1. *A simple ring A is left primitive with faithful simple left A-module $A/\mathfrak{m}$ for any left maximal ideal $\mathfrak{m}$ of A.*

2. *A commutative ring A is primitive if and only if A is a field.*

§1.2 Jacobson Density Theorem

Definition 1.7. *Let V be a vector space over a division ring D. A subring A of $\operatorname{End}_D(V)$ is called a* ***dense ring of endomorphisms*** *of V if for every positive integer n, every linearly independent subset $\{u_1, \ldots, u_n\}$ of V and every arbitrary subset $\{v_1, \ldots, v_n\}$ of V, there exists $f \in A$ such that $f(u_i) = v_i, (i = 1, 2, \ldots, n)$.*

Lemma 1.8. *Let M be a simple module over a ring A. Consider M as a vector space over the division ring $D = \operatorname{End}_A(M)$. If V is a finite dimensional D-subspace of M and $a \in M - V$, then there exists $r \in A$ such that $ra \neq 0$ and $rV = 0$.*

Remark. *In other words, the element $r \in \operatorname{Ann}_A(V)$ only annihilates D-subspace V.*

Proof. The proof is by induction on $n = \dim_D V$. If $n = 0$, then $V = 0$ and $a \neq 0$. Since M is simple, $M = Ra$. Consequently, there exists $r \in R$ such that $ra = a \neq 0$ and $rV = r0 = 0$.

Suppose now $\dim_D V = n > 0$ and the theorem is true for dimensions less than n. Let $\{u_1, \ldots, u_{n-1}, u\}$ be a D-basis of V and let $W = \operatorname{span}\{u_1, \ldots, u_{n-1}\}$ ($W = 0$ if $n = 1$). Then $V = W \oplus Du$ (vector space direct sum, W may not be an R-submodule of M) the left annihilator $I = \operatorname{Ann}_R(W)$ is a left ideal of R.

Consequently, Iu is an R-submodule of M. Since $u \in M - W$, the induction hypothesis implies that there exists $r \in R$ such that $ru \neq 0$ and $rW = 0$. Consequently $r \in I$ and $0 \neq ru \in Iu$, whence $Iu \neq 0$. Therefore $M = Iu$ by simplicity.

We must find $r \in R$ such that $ra \neq 0$ and $rV = 0$. If no such r exists, $\operatorname{Ann}(a) \subset \operatorname{Ann}(V)$, then we can define a map $\theta : M \to M$ as follows. For $ru \in Iu = M$ let $\theta(ru) = ra \in M$. We claim that θ is well defined. If $r_1 u = r_2 u\,(r_i \in I)$, then $(r_1 - r_2)\, u = 0$, whence $(r_1 - r_2)\, V$ $= (r_1 - r_2)(W \oplus Du) = 0$. Consequently by hypothesis $(r_1 - r_2)\, a = 0$. Therefore, $\theta(r_1 u) =$ $r_1 a = r_2 a = \theta(r_2 u)$. Verify that $\theta \in \operatorname{Hom}_R(M, M) = D$. Then for every $r \in I$,

$$0 = \theta(ru) - ra = r\theta(u) - ra = r(\theta(u) - a)$$

Therefore $\theta(u) - a \in W$ by induction hypothesis. Consequently

$$a = \theta u - (\theta u - a) \in Du + W = V,$$

which contradicts the fact that $a \notin V$. Therefore, there exists $r \in R$ such that $ra \neq 0$ and $rV = 0$. □

Theorem 1.9 (Classic Jacobson Density Theorem)**.** *Let A be a left primitive ring with a faithful simple left A-module M and division ring $D = \mathrm{End}_A(M)$. Then A is a dense ring of endomorphisms of the D-vector space M (viewed $\alpha : A \hookrightarrow \mathrm{End}_D(M)$ by $a \mapsto \alpha_a$ where left regular action $\alpha_a : m \mapsto am$ in M).*

Proof. It clear that $\alpha : A \to \mathrm{End}_D(M)$ is a ring monomorphism since M is faithful. Let $\{u_1, u_2, \ldots, u_n\}$ be a D-linearly independent subset and $\{v_1, v_2, \ldots, v_n\}$ be an arbitrary subset. For each i let

$$V_i = \mathrm{span}\left\{u_1, \ldots, u_{i-1}, u_{i+1}, \ldots, u_n\right\}.$$

Since U is D-linearly independent, $u_i \notin V_i$. Consequently, by 1.8 there exists $r_i \in R$ such that

$$r_i u_i \neq 0 \text{ and } r_i V_i = 0$$

whence $R r_i u_i = M$ by simplicity. Therefore exists $t_i \in R$ such that $t_i r_i u_i = v_i$. Let

$$r = t_1 r_1 + t_2 r_2 + \cdots + t_n r_n \in R.$$

Consequently for each $i = 1, 2, \ldots, n$

$$\alpha_r\left(u_i\right) = \left(t_1 r_1 + \cdots + t_n r_n\right) u_i = v_i$$

Therefore $\mathrm{Im}\,\alpha$ is a dense ring of endomorphisms of the D-vector space M. □

Corollary 1.10. *If M is a simple left A-module, then A acts densely on M. That is, for every positive integer n, every linearly independent subset $\{u_1, \ldots, u_n\}$ of M and every arbitrary subset $\{v_1, \ldots, v_n\}$ of M, there exists $a \in A$ such that $au_i = v_i$, $(i = 1, 2, \ldots, n)$.*

§1.3 Artinian simple ring

Lemma 1.11. *Let A be a dense ring of endomorphisms of a vector space V over a division ring D. Then A is left Artinian if and only if $\dim_D V$ is finite, in which case $A = \mathrm{End}_D(V) \cong M_n(D)$.*

Proof. If A is Artinian and $\dim_D V$ is infinite, then there exists an infinite linearly independent subset $\{u_1, u_2, \ldots\}$ of V. The V is a left simple $\mathrm{End}_D(V)$-module and hence a left simple A-module.

For each n, let $I_n = \mathrm{Ann}_A\{u_1, \ldots, u_n\}$. Then $I_1 \supsetneq I_2 \supsetneq \cdots$ is a properly descending chain by lemma 1.8 , which is a contradiction. Hence $\dim_D V$ is finite.

Conversely if $\dim_D V$ is finite, then V has a finite basis $\{v_1, \ldots, v_m\}$. Then $A = \mathrm{Hom}_D(V, V) \cong M_n(D)$ is Artinian. □

Corollary 1.12 (Artin-Wedderburn). *If A is a simple ring, then*

1. *A is left Artinian*
2. *$A \cong M_n(D)$ for some positive integer n and some division ring $D \cong \mathrm{End}_A(A/\mathfrak{m})$ for some maximal left ideal $\mathfrak{m}$ of A.*
3. *A is right Artinian*

Lemma 1.13 (Schur's lemma). *Let V be a nonzero vector space over a division ring D. If $g : V \to V$ is a homomorphism of additive groups such that $gf = fg$ for all $f \in \mathrm{Hom}_D(V, V)$,then there exists $\lambda \in D$ such that $g(x) = \lambda x$ for all $x \in V$.*

Remark. *That is, that is,*

$$g \in \mathrm{End}_{\mathrm{End}_D(V)}(V) \cong D$$

since V is a simple $\mathrm{End}_D(V)$-module.

Lemma 1.14. *The simple left $M_n(D)$-modules for some division ring D is isomorphic to D^n with the usual left matrix multiplication.*

Proposition 1.15. *If $M_{n_1}(D_1) \cong M_{n_2}(D_2)$ for some positive integers n_1, n_2 and some division rings D_1, D_2, then $n_1 = n_2$ and $D_1 \cong D_2$.*

Proof.

$$D_1 \cong \mathrm{End}_{M_{n_1}(D_1)}(D_1^{n_1}) \cong \mathrm{End}_{M_{n_2}(D_2)}(D_2^{n_2}) \cong D_2$$

□

§2 Semisimplicity

Theorem 2.1. *Let A be a ring and M a left A-module. The following conditions on M are equivalent:*

1. *M is the sum of a family of simple submodules.*
2. *M is the direct sum of a family of simple submodules.*
3. *Every submodule N is a direct summand of M.*

Proof. (1) ⇒ (2) Let $\mathcal{S}$ be the set of all families $\mathcal{F}$ of simple submodules of M such that the sum of the members of $\mathcal{F}$ is direct. Since M is the sum of a family of simple submodules, $\mathcal{S}$ is nonempty. Partially order $\mathcal{S}$ by inclusion and let $\mathcal{C}$ be a chain in $\mathcal{S}$. Then $\mathcal{U} = \bigcup_{\mathcal{F} \in \mathcal{C}} \mathcal{F}$ is an upper bound for

$\mathcal{C}$ in $\mathcal{S}$. By Zorn's lemma there exists a maximal element $\mathcal{F}_0$ in $\mathcal{S}$. We claim that $M = \bigoplus_{N \in \mathcal{F}_0} N$. If not, there exists a simple submodule K of M such that

$$K \cap \left(\bigoplus_{N \in \mathcal{F}_0} N \right) = 0.$$

Consequently, $\mathcal{F}_0 \cup \{K\} \in S$, contradicting the maximality of $\mathcal{F}_0$.

(2) $\Rightarrow$ (3) Let $M = \bigoplus_{i \in I} N_i$, where each N_i is a simple submodule of M, and let N be a submodule of M. For each $i \in I$, either $N_i \subset N$ or $N_i \cap N = 0$ by simplicity. Let $J = \{i \in I \mid N_i \subset N\}$ and $K = I - J$. Then

$$M = N \oplus \left(\bigoplus_{i \in K} N_i \right).$$

(3) $\Rightarrow$ (1) Let N be the sum of all simple submodules of M. By hypothesis, $M = N \oplus P$ for some submodule □

Remark. *A left A-module M satisfying the three conditions is said to be* ***semisimple****. Similarly one defines a right semisimple module.*

Proposition 2.2. *Every submodule and every factor module of a left semisimple module is left semisimple.*

§2.1 Structure of semisimple rings

Definition 2.3. *A ring A is called* ***semisimple*** *if $1_A \neq 0$, and if ${}_AA$ is a semisimple module.*

Proposition 2.4. *If A is a semisimple ring, then A is left Artinian.*

Proof. Since ${}_AA$ is semisimple module, there exist simple left ideals L_j of A such that

$${}_AA \cong \bigoplus_{j \in J} L_j$$

with projection maps $\pi_j : {}_AA \to {}_AL_j$ for each $j \in J$. Since the sum is direct, $\{j_k : \pi_{j_k}(1_A) \neq 0\}$ is finite. Then $J = \{j_k : \pi_{j_k}(1_A) \neq 0\}$ is finite and

$${}_AA \cong L_1 \oplus L_2 \oplus \cdots \oplus L_n$$

Therefore ring A is left Artinian. □

Remark. *It is clear that any simple left ideal L of A is isomorphic to some L_j, thus A has finitely many minimal left ideals up to isomorphism.*

Lemma 2.5. *If L and L' are minimal left ideals in a ring R, then each of the following statements implies the one below it:*

1. $LL' \neq (0)$.

2. $\mathrm{Hom}_R(L, L') \neq \{0\}$ *and there exists* $b' \in L'$ *with* $L' = Lb'$.

3. $L \cong L'$ *as left* R*-modules.*

If also $L^2 \neq (0)$*, then (iii) implies (i) and the three statements are equivalent.*

Theorem 2.6 (Wedderburn-Artin)**.** *If* A *is a semisimple ring, then we have ring isomorphisms*

$$A \cong \prod_{i=1}^{t} A_i \cong \prod_{i=1}^{t} M_{n_i}(D_i)$$

where $A_i \cong M_{n_i}(D_i)$ $i \in I$.

Proof. Let $\{L_{j_i}\}_{i=1}^{t}$ be all distinct simple left ideals of A up to isomorphism and define A_i be the sum

$$_AA_i := \bigoplus_{L_j \cong L_i} L_j$$

Thus, as left A-modules,

$$_AA \cong \bigoplus_{i=1}^{t} {}_AA_i.$$

Then A_i is a two-sided ideal of A by lemma 2.5 and hence a ring with identity element $e_i = \pi_i(1_A)$ (π_i the projection from A to A_i). The projection from A to A_i induce a ring homomorphism

$$A \cong \prod_{i=1}^{t} A_i$$

since $1_A = e_1 + e_2 + \cdots + e_t$ and $e_ie_j = \delta_{ij}1_A$.

Finally, each A_i is a simple ring since any nonzero two-sided ideal $\mathfrak{a}$ of A_i contains a minimal left ideal L of A_i and hence a minimal left ideal of A isomorphic to L_i. Then $Lb \subset L$ runs over all minimal left ideals of A isomorphic to L_i by lemma 2.5 , whence $A_i \subset L \subset \mathfrak{a}$. □

§2.2 Structure of modules over semisimple rings

Theorem 2.7. *Let semisimple ring*

$$A \cong \prod_{i=1}^{t} A_i$$

where each $A_i \cong M_{n_i}(D_i)$*. Then every left* A*-module* M *is isomorphic to*

$$\bigoplus_{i=1}^{t} e_iM \cong \bigoplus_{i=1}^{t} S_i^{(\kappa_i)}$$

where $e_iM \cong S_i \cong D_i^{n_i}$ *and* κ_i *is a cardinal number.*

And the decomposition is unique in the sense that if

$$M \cong \bigoplus_{i=1}^{t} S_i^{(\kappa_i)} \cong \bigoplus_{i=1}^{t} S_i^{\left(\kappa_i'\right)}$$

then $\kappa_i = \kappa_i'$ for each i

Remark. *If M is finitely generated, then each κ_i is finite.*

§3 Jacobson Radical

Definition 3.1. *Let R be a ring. A element $x \in R$ is said to be **right quasi-regular** if there exists $y \in R$ such that $x + y - xy = 0$, y is called a **right quasi-inverse** of x.*

Remark. *That is, $1 - x$ has a right inverse $1 - y$.*

Definition 3.2. *A element $a \in R$ is said **right quasi-nilpotent element** if for every $r \in R$, ra is right quasi-regular.*

Remark. *That is, $1 - ra$ has a right inverse for every $r \in R$.*

§3.1 Jacobson radical

Theorem 3.3. *Let R be a ring, then the following sets are equal:*

1. *the intersection of all maximal left ideals of R*

1' *the intersection of all maximal right ideals of R*

2. *the intersection of all the annihilators of simple left R-modules;*

2' *the intersection of all the annihilators of simple right R-modules;*

3. $\{x \in R : x$ *is right quasi-nilpotent element*$\}$

3' $\{x \in R : x$ *is left quasi-nilpotent element*$\}$

4. *Nakayama's* $\{x \in R :$ *for any finitely generated module* $M, xM = M \Rightarrow M = 0\}$

5. *the ideal J such that R/J is semisimple.*

*the two-sided ideal is called the **Jacobson radical** of R and is denoted by $J(R)$.*

Corollary 3.4. 1. $J\left(\prod_{i\in I} R_i\right) = \prod_{i\in I} J\left(R_i\right)$.

2. $J\left(J\left(R\right)\right) = J\left(R\right)$.

3. *$R/J(R)$ is semisimple.*

Proposition 3.5. *If R is a Artinian ring, then the radical $J(R)$ is a nilpotent ideal. Thus $J(R) =$ Rad (R)*

Proof. Since R is Artinian, the descending chain of ideals

$$J(R) \supseteq J(R)^2 \supseteq J(R)^3 \supseteq \cdots$$

stabilizes, say $J(R)^n = J(R)^{n+1}$ for some positive integer n. On the other hand, ${}_RR$ is Noetherian thus $J(R)^n$ is finitely gengerated submodule of ${}_RR$ and $J(R)^n = 0$ by Nakayama's lemma. □

Proposition 3.6. *Let R be a ring. $J(M_nR) = M_n(J(R))$.*

Proof. (a) If A is a left R-module, consider the elements of $A^n = A \oplus A \oplus \cdots \oplus A$ (n summands) as column vectors; then A^n is a left $(\mathrm{Mat}_n R)$-module (under ordinary matrix multiplication).

(b) If A is a simple R-module, A^n is a simple $(\mathrm{Mat}_n R)$-module.

(c) $J(\mathrm{Mat}_n R) \subset \mathrm{Mat}_n J(R)$.

(d) $\mathrm{Mat}_n J(R) \subset J(\mathrm{Mat}_n R)$. [Hint: prove that $\mathrm{Mat}_n J(R)$ is a left quasi-regular ideal of $\mathrm{Mat}_n R$ as follows. For each $k = 1, 2, \ldots, n$ let K_k consist of all matrices (a_{ij}) such that $a_{ij} \varepsilon J(R)$ and $a_{ij} = 0$ if $j \neq k$. Show that K_k is a left quasi-regular left ideal of $\mathrm{Mat}_n R$ and observe that $K_1 + K_2 + \cdots + K_n = \mathrm{Mat}_n J(R)$.] □

Corollary 3.7. *Let R be a ring. If matrix $A \in M_n(J(R))$, then $I - A$ is invertible in $M_n(R)$.*

§3.2 Nakayama's Lemma

Theorem 3.8 (Nakayama's lemma)**.** *Let M be a finitely generated left R-module and a subset $X \subset J(R)$. Then $\mathfrak{a}M = M$ implies $M = 0$.*

Corollary 3.9. *Let M be a finitely generated left R-module, N a submodule of M, left ideal $\mathfrak{a} \subset J(R)$. Then $M = \mathfrak{a}M + N \Rightarrow M = N$.*

Corollary 3.10. *If R is a Noetherian local ring with maximal ideal $\mathfrak{m}$, then $\bigcap_{n=1}^{\infty} \mathfrak{m}^n = 0$.*

§4 Characterizations of semisimple rings

Theorem 4.1. *The following conditions on a ring R are equivalent:*

1. *R is a semisimple ring.*
2. *Every left R-module is a semisimple module.*
3. *Every left R-module is injective.*
4. *Every left R-module is projective.*
5. *Every short exact sequence of left R-modules splits.*
6. *R is Artinian and $J(R) = 0$.*

§5 Algebra

Definition 5.1. *Let A be an algebra over a commutatuve ring K with identity.*

1. *A **left algebra** A**-module** is a left K-module M such that M is a left module over the ring A and $k(am) = (ka)m = a(km)$ for all $k \in K, a \in A, m \in M$. Indeed,*

$$\begin{cases} (k_1a_1 + k_2a_2)(m_1 + m_2) = k_1a_1m_1 + k_1a_1m_2 + k_2a_2m_1 + k_2a_2m_2 \\ k(am) = (ka)m = a(km) \\ 1_K m = m, 1_K a = a \end{cases}$$

   for all $k \in K, a \in A, m \in M$

2. *A left algebra A-submodule of M is a subset of M which is itself an left algebra A-module.*

3. *A left algebra A-module M is **simple** (or **irreducible**) if M has no proper A-submodules.*

4. *A homomorphism $f : M \to N$ of algebra A-modules is a map that is both a K-module and an A-module homomorphism.*

Remark.

Theorem 5.2. *Let A be a K-algebra. The Jacobson radical of the ring A coincides with the Jacobson radical of the algebra A. In particular A is a semisimple ring if and only if A is a semisimple algebra.*

Theorem 5.3. *Let A be a K-algebra.*

(1) Every simple algebra A-module is a simple module over the ring A.

(2) Every simple module M over the ring A can be given a unique K-module structure in such a way that M is a simple algebra A-module.

Chapter VIII

Integral

§1 Rings Extensions

Definition 1.1. *Let B be a commutative ring with identity and A a subring of B containing 1_B.*

1. *Then B is said to be an* ***extension ring*** *of A.*
2. *An element $s \in B$ is said to be* ***integral*** *over A if s is a root of a monic polynomial in $A[x]$.*
3. *If every element of B is integral over A, B is said to be an* ***integral extension*** *of A.*
4. *The* ***integral closure*** *of A in B is the set of elements of B that are integral over A.*
5. *The ring A is said to be* ***integrally closed*** *in B if A is equal to its integral closure in B.*

   The integral closure of an integral domain R in its field of fractions is called the ***normalization*** *of R. An integral domain is called integrally closed or normal if it is integrally closed in its field of fractions.*

Remark. *If follows from corollary 1.3 that the integral closure of R in S is a subring of S containing R.*

Theorem 1.2. *Let B be an extension ring of A and $s \in B$. Then the following conditions are equivalent.*

1. *s is integral over A*
2. *Subring $A[s]$ is a finitely generated A-module*
3. *There is a subring T that $A[s] \subset T \subset B$, which is finitely generated as an A-module;*
4. *There is a faithful $A[s]$-submodule M which is finitely generated as an A-module.*

Corollary 1.3. *Let B be an extension ring of A. Then*

1. *If B is finitely generated as an A-module, then B is an integral extension of A.*

2. *If $s_1, \ldots, s_t \in B$ are integral over A, then $A[s_1, \ldots, s_t]$ is a finitely generated A-module and an integral extension ring of A.*

3. *If T is an integral extension ring of B and B is an integral extension ring of A, then T is an integral extension ring of A .*

Proof. It immediately follows from 1.2 □

Proof. We have a tower of extension rings:

$$R \subset R[s_1] \subset R[s_1, s_2] \subset \cdots \subset R[s_1, \ldots, s_t]$$

For each i, s_i is integral over R and hence integral over $R[s_1, \ldots, s_{i-1}]$. Since $R[s_1, \ldots, s_i] = R[s_1, \ldots, s_{i-1}][s_i], R[s_1, \ldots, s_i]$ is a finitely generated module over $R[s_1, \ldots, s_{i-1}]$ by 1.2. Thus $R[s_1, \ldots, s_n]$ is a finitely generated R-module, then $R[s_1, \ldots, s_n]$ is an integral extension ring of R by 1. □

Proof. T is obviously an extension ring of R. If $t \in T$, then t is integral over S and therefore the root of some monic polynomial $f \in S[x]$, say $f = \sum_{i=0}^{n} s_i x^i$. Since f is also a polynomial over the ring $R[s_0, s_1, \ldots, s_{n-1}], t$ is integral over $R[s_0, \ldots, s_{n-1}]$.

By 1.2 $R[s_0, \ldots, s_{n-1}][t]$ is a finitely generated $R[s_0, \ldots, s_{n-1}]$-module. But since S is integral over $R, R[s_0, \ldots, s_{n-1}]$ is a finitely generated R-module by 2. Then

$$R[s_0, \ldots, s_{n-1}][t] = R[s_0, \ldots, s_{n-1}, t]$$

is a finitely generated R-module. Since $R[t] \subset R[s_0, \ldots, s_{n-1}, t], t$ is integral over R by 1.2. □

Proposition 1.4. 1. *Every unique factorization domain is integrally closed.*

2. *In particular, the polynomial ring $F[x_1, \ldots, x_n]$ (F a field) is integrally closed in its quotient field $F(x_1, \ldots, x_n)$.*

§1.1 Integral extension

Theorem 1.5. *Let T be a multiplicative subset of an integral domain R such that $0 \notin T$. If R is integrally closed, then $T^{-1}R$ is an integrally closed integral domain.*

Proof. $T^{-1}R$ is an integral domain and R may be identified with a subring of $T^{-1}R$ by 1.2. Extending this identification, the quotient field $Q(R)$ of R may be considered as a subfield of the quotient field $Q(T^{-1}R)$ of $T^{-1}R$. Verify that $Q(R) = Q(T^{-1}R)$.

Let $u \in Q(T^{-1}R)$ be integral over $T^{-1}R$; then for some $r_i \in R$ and $s_i \in T$,

$$u^n + (r_{n-1}/s_{n-1}) u^{n-1} + \cdots + (r_1/s_1) u + (r_0/s_0) = 0.$$

Multiply through this equation by s^n, where $s = s_0 s_1 \cdots s_{n-1} \in T$, and conclude that su is integral over R. Since $su \in Q\left(T^{-1}R\right) = Q(R)$ and R is integrally closed, $su \in R$. Therefore, $u = su/s \in T^{-1}R$, whence $T^{-1}R$ is integrally closed. $\square$

Theorem 1.6. *Let S be an integral extension ring of R. Then the following statements hold.*

1. *Assume that S is an integral domain. Then R is a field if and only if S is a field.*

2. *Let $\mathfrak{p}$ be a prime ideal in R. Then there is a prime ideal $\mathfrak{q}$ in S with $\mathfrak{p} = \mathfrak{q} \cap R$.*

   Moreover, $\mathfrak{p}$ is maximal if and only if $\mathfrak{q}$ is maximal.

3. *(The Going-up Theorem) Let $\mathfrak{p}_1 \subseteq \mathfrak{p}_2 \subseteq \cdots \subseteq \mathfrak{p}_n$ be a chain of prime ideals in R and suppose there are prime ideals $\mathfrak{q}_1 \subseteq \mathfrak{q}_2 \subseteq \cdots \subseteq \mathfrak{q}_m$ of S with $\mathfrak{p}_i = \mathfrak{q}_i \cap R, 1 \leq i \leq m$ and $m < n$. Then the ascending chain of ideals can be completed: there are prime ideals $\mathfrak{q}_{m+1} \subseteq \cdots \subseteq \mathfrak{q}_n$ in S such that $\mathfrak{p}_i = \mathfrak{q}_i \cap R$ for all i.*

Theorem 1.7 (The Going-down Theorem)**.** *Assume that S is an integral domain and R is integrally closed in S. Let $\mathfrak{p}_1 \supseteq \mathfrak{p}_2 \supseteq \cdots \supseteq \mathfrak{p}_n$ be a chain of prime ideals in R and suppose there are prime ideals $\mathfrak{q}_1 \supseteq \mathfrak{q}_2 \supseteq \cdots \supseteq \mathfrak{q}_m$ of S with $\mathfrak{p}_i = \mathfrak{q}_i \cap R, 1 \leq i \leq m$ and $m < n$. Then the descending chain of ideals can be completed: there are prime ideals $\mathfrak{q}_{m+1} \supseteq \cdots \supseteq \mathfrak{q}_n$ in S such that $\mathfrak{p}_i = \mathfrak{q}_i \cap R$ for all i.*

Theorem 1.8. *Let S be an integral extension ring of R and let $\mathfrak{q}$ be a prime ideal in S which lies over a prime ideal $\mathfrak{p}$ in R . Then $\mathfrak{q}$ is maximal in S if and only if $\mathfrak{p}$ is maximal in R .*

Proof. Suppose $\mathfrak{q}$ is maximal in S. By **??** there is a maximal ideal $\mathfrak{m}$ of R that contains $\mathfrak{p}$ and $\mathfrak{m}$ is prime by **??**. By **??** there is a prime ideal $\mathfrak{q}'$ in S such that $\mathfrak{q} \subset \mathfrak{q}'$ and $\mathfrak{q}'$ lies over $\mathfrak{m}$. Since $\mathfrak{q}'$ is prime, $\mathfrak{q}' \neq S$. The maximality of $\mathfrak{q}$ implies that $\mathfrak{q} = \mathfrak{q}'$, whence $\mathfrak{p} = \mathfrak{q} \cap R = \mathfrak{q}' \cap R = \mathfrak{m}$. Therefore, $\mathfrak{p}$ is maximal in R.

Conversely suppose $\mathfrak{p}$ is maximal in R. Since $\mathfrak{q}$ is prime in $S, \mathfrak{q} \neq S$ and there is a maximal ideal N of S containing $\mathfrak{q}$ and N is prime, whence $1_R = 1_S \notin N$. Since $\mathfrak{p} = R \cap \mathfrak{q} \subset R \cap N \subset R$, we must have $\mathfrak{p} = R \cap N$ by maximality. Thus $\mathfrak{q}$ and N both lie over $\mathfrak{p}$ and $\mathfrak{q} \subset N$. Therefore, $\mathfrak{q} = N$ by 1.8 . $\square$

§2 Discrete Valuation Ring

Definition 2.1. *The following conditions on a principal ideal domain are equivalent:*

1. *A has exactly one nonzero prime ideal;*

2. *up to associates, A has exactly one prime element;*

3. *A is local and is not a field.*

A ring satisfying these conditions is called a ***discrete valuation ring****.*

Theorem 2.2. *An integral domain A is a discrete valuation ring if and only if*

(i) A is noetherian,

(ii) A is integrally closed, and

(iii) A has exactly one nonzero prime ideal.

§3 Dedekind Domain

Definition 3.1. *A **Dedekind domain** is an integral domain R satisfying the following equivalent conditions:*

1. *R is Noetherian, integrally closed and has Krull dimension one (Every nonzero prime ideal of R is maximal).*
2. *Every nonzero ideal of R is invertible*
3. *Every finitely generated torsion-free R-module is free.*
4. *the localization $R_{\mathfrak{p}}$ at each prime ideal $\mathfrak{p}$ of R is a discrete valuation ring.*
5. *Every nonzero proper ideal of R can be written as a product of prime ideals of R, and this factorization is unique up to the order of the factors.*

Proposition 3.2. *Let A be an integral domain, and let S be a multiplicative subset of A.*

1. *If A is noetherian, then so also is $S^{-1}A$.*
2. *If A is integrally closed, then so also is $S^{-1}A$.*
3. *If A has Krull dimension one, then so also does $S^{-1}A$.*
4. *If A is a Dedekind domain, then so also is $S^{-1}A$.*

Proposition 3.3. *A noetherian integral domain A is a Dedekind domain if and only if, for every nonzero prime ideal $\mathfrak{p}$ in A, the localization $A_{\mathfrak{p}}$ is a discrete valuation ring.*

§3.1 Unique factorization of ideals

Theorem 3.4. *Let A be a Dedekind domain. Every proper nonzero ideal $\mathfrak{a}$ of A can be written in the form*

$$\mathfrak{a} = \mathfrak{p}_1^{r_1} \cdots \mathfrak{p}_n^{r_n}$$

with the $\mathfrak{p}_i$ distinct prime ideals and the $r_i > 0$; the $\mathfrak{p}_i$ and the r_i are uniquely determined.

§3.2 The ideal class group

Definition 3.5. *Let R be an integral domain with quotient field K. A* ***fractional ideal*** *of R is*

(i) a nonzero R-submodule I of K

(ii) there exists a nonzero $d \in R$ such that $dI \subset R$ i,e, $(R : I) \cap R \neq \varnothing$

Definition 3.6. *If R is an integral domain with quotient field K, then the set of all fractional ideals of R forms a commutative monoid, with identity R and multiplication given by*

$$IJ = \left\{ \sum_{i=1}^{n} a_i b_i \mid a_i \in I; b_i \in J; n \in \mathbb{Z}_{\geq 1} \right\}$$

A fractional ideal I of an integral domain R is said to be ***invertible*** *if $IJ = R$ for some fractional ideal J of R.*

Theorem 3.7. *Let A be a Dedekind domain. The set* $\mathrm{Id}(A)$ *of fractional ideals is a group; in fact, it is the free abelian group on the set of nonzero prime ideals.*

Definition 3.8. *We define the* ***ideal class group*** $\mathrm{Cl}(A)$ *of A to be the quotient* $\mathrm{Cl}(A) = \mathrm{Id}(A)/\mathrm{P}(A)$ *of* $\mathrm{Id}(A)$ *by the subgroup of principal ideals. The* ***class number*** *of A is the order of* $\mathrm{Cl}(A)$ *(when finite).*

In the case that A is the ring of integers $\mathcal{O}_K$ in a number field K, we often refer to $\mathrm{Cl}(\mathcal{O}_K)$ *as the ideal class group of K, and its order as the class number of K.*

Proposition 3.9. *Let R be an integral domain with quotient field K.*

1. *Indeed for any fractional ideal I the set $I^{-1} = \{a \in K \mid aI \subset R\}$ is easily seen to be a fractional ideal such that $I^{-1}I = II^{-1} \subset R$.*

2. *The inverse of an invertible fractional ideal I is unique and is $I^{-1} = \{a \in K \mid aI \subset R\}$. If I is invertible and $IJ = JI = R$, then clearly $J \subset I^{-1}$. Conversely, since I^{-1} and J are R-submodules of $K, I^{-1} = RI^{-1} = (JI)I^{-1} = J(II^{-1}) \subset JR = RJ \subset J$, whence $J = I^{-1}$.*

3. *If I, A, B are fractional ideals of R such that $IA = IB$ and I is invertible, then $A = RA = (I^{-1}I)A = I^{-1}(IB) = RB = B$.*

4. *If I is an ordinary ideal in R, then $R \subset I^{-1}$.*

Lemma 3.10. *Let $I, I_1, I_2, \ldots, I_n$ be ideals in an integral domain R .*

1. *The ideal $I_1 I_2 \cdots I_n$ is invertible if and only if each I_j is invertible.*

2. *If $\mathfrak{p}_1 \cdots \mathfrak{p}_m = I = \mathfrak{q}_1 \cdots \mathfrak{q}_n$, where the $\mathfrak{p}_i$ and $\mathfrak{q}_j$ are prime ideals in R and every $\mathfrak{p}_i$ is invertible, then $m = n$ and (after reindexing) $\mathfrak{p}_i = \mathfrak{q}_i$ for each $i = 1, \ldots, m$.*

Lemma 3.11. *If I is a fractional ideal of an integral domain R with quotient field K and $f \in \mathrm{Hom}_R(I, R)$, then for all $a, b \in I : af(b) = bf(a)$.*

Lemma 3.12. *Every invertible fractional ideal of an integral domain R with quotient field K is a finitely generated R-module.*

Theorem 3.13. *Let R be an integral domain and I a fractional ideal of R. Then I is invertible if and only if I is a projective R-module.*

§4 Discrete valuations

Definition 4.1. *Let K be a field. A* ***discrete valuation*** *on K is a nonzero homomorphism $v : K^\times \to \mathbb{Z}$ such that $v(a+b) \geq \min(v(a), v(b))$.*

As v is not the zero homomorphism, its image is a nonzero subgroup of $\mathbb{Z}$, and is therefore of the form $m\mathbb{Z}$ for some $m \in \mathbb{Z}$. If $m = 1$, then $v : K^\times \to \mathbb{Z}$ is surjective, and v is said to be ***normalized****; otherwise, $x \mapsto m^{-1} \cdot v(x)$ will be a normalized discrete valuation.*

We extend v to a map $K \to \mathbb{Z} \cup \{\infty\}$ by setting $v(0) = \infty$, where ∞ is a symbol $\geq n$ for all $n \in \mathbb{Z}$.

Remark. *We have*

1. *$v(\zeta) = 0$ for some $\zeta \in K^\times$*
2. *$v(-a) = v(a)$ for all $a \in K$;*
3. *$v(a+b) = \max\{v(a), v(b)\}$ if $v(a) \neq v(b)$.*

We often use "ord" rather than " v " to denote a discrete valuation.

Proposition 4.2. *Let v be a discrete valuation on K, then*

$$A := \{a \in K \mid v(a) \geq 0\}$$

is a principal ideal domain with maximal ideal

$$\mathfrak{m} = \{a \in K \mid v(a) > 0\}$$

If $v(K^\times) = m\mathbb{Z}$, then the ideal $\mathfrak{m}$ is generated by every element of $v^{-1}(m)$.

Definition 4.3. *Let A be a Dedekind domain and let $\mathfrak{p}$ be a prime ideal in A. For any $c \in K^\times$, let $v(c)$ be the exponent of $\mathfrak{p}$ in the factorization of (c). Then v is a normalized discrete valuation on K, called the* ***discrete valuation associated to*** *$\mathfrak{p}$, denoted by $\mathrm{ord}_{\mathfrak{p}}$.*

Proposition 4.4. *Let $x_1, \ldots, x_m$ be elements of a Dedekind domain A, and let $\mathfrak{p}_1, \ldots, \mathfrak{p}_m$ be distinct prime ideals of A. For every integer n, there is an $x \in A$ such that*

$$\mathrm{ord}_{\mathfrak{p}_i}(x - x_i) > n, \quad i = 1, 2, \ldots, m.$$

§5 Integral closures of Dedekind domains

Theorem 5.1. *Let A be a Dedekind domain with field of fractions K and L/K be a finite separable extension, then the integral closure of A in L is Dedekind domain.*

Definition 5.2. *Let A be a Dedekind domain with field of fractions K, and let B be the integral closure of A in a finite separable extension L of K. A prime ideal $\mathfrak{p}$ of A will factor in B,*

$$\mathfrak{p}B = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_g^{e_g}$$

wher $\mathfrak{P}$ are distinct prime ideals in B and $e_i \geq 1$,

1. *If any of the numbers is > 1, then we say that $\mathfrak{p}$ is* ***ramified*** *in B (or L). The number e_i is called the* ***ramification index*** *.*

2. *We say $\mathfrak{P}$ divides $\mathfrak{p}$,written $\mathfrak{P} \mid \mathfrak{p}$, if $\mathfrak{P}$ occurs in the factorization of $\mathfrak{p}$ in B.*

We then write $e(\mathfrak{P}/\mathfrak{p})$ for the ramification index and $f(\mathfrak{P}/\mathfrak{p})$ for the degree of the field extension $[B/\mathfrak{P} : A/\mathfrak{p}]$ (called the ***residue class degree****).*

3. *$\mathfrak{p}$ is said to* ***split*** *(or split completely) in L if $e_i = f_i = 1$ for all i*

4. *$\mathfrak{p}$ is said to be* ***inert*** *in L if $\mathfrak{p}B$ is a prime ideal (so $g = 1 = e$).*

Theorem 5.3. *Let m be the degree of L over K, and let $\mathfrak{P}_1, \ldots, \mathfrak{P}_g$ be the prime ideals dividing $\mathfrak{p}$; then*

$$\sum_{i=1}^{g} e_i f_i = m$$

where $e_i = e\left(\mathfrak{P}_i/\mathfrak{p}\right)$ and $f_i = f\left(\mathfrak{P}_i/\mathfrak{p}\right)$. If L is Galois over K, then all the ramification numbers are equal, and all the residue class degrees are equal, and so

$$efg = m.$$

Chapter IX

Completions

§1

§2 Graded rings and graded modules

§2.1 Graded ring

Definition 2.1. *A* ***graded ring*** *is a ring A together with a family $\{A_n\}_{n\geq 0}$ of subgroups of A, such that*

(i) $A = \bigoplus_{n=0}^{\infty} A_n$ *in* **Ab**

(ii) *and $A_m A_n \subseteq A_{m+n}$ for all $m, n \geq 0$.*

Thus A_0 must be a subring of A containing 1_A, and A is a A_0-algebra. A is saied to be ***standard graded ring*** *if*

(iii) $A = A_0[A_1]$ *as a A_0-algebra.*

Proposition 2.2. *Let $A = \bigoplus_{n\geq 0} A_n$ be a graded ring. Then the following statements are equivalent:*

1. *A is a Notetherian ring.*
2. *A_0 is Noetherian and A is finitely generated as A_0-algebra (A_0-ring)*

Definition 2.3. *Let $A = \bigoplus A_n$ be a graded ring.*

1. *A ideal (left, right, two-sided) I of A is called* ***graded ideal*** *if $I = \bigoplus_{n\geq 0}(A_n \cap I)$ in* **Ab**.
2. *If I is a two-sided graded ideal of A, then quotient graded ring $A/I = \bigoplus_{i\geq 0} A_i/\left(A_i \cap I\right)$ is well-defined, called the* ***quotient graded ring*** *of A by I.*

Remark. *It clear that a ideal I is graded if and only if it is generated by homogeneous elements.*

Definition 2.4. *Let A and B be graded rings.*

1. *A **graded ring homomorphism** from A to B is a ring homomorphism $f : A \to B$ such that $f(A_n) \subseteq B_n$ for all $n \geqslant 0$.*

2. *the **kernel** of a graded ring homomorphism is a two-sided graded ideal.*

§2.2 Graded modules

Definition 2.5. *Let $A = \bigoplus_{n\geq 0} A_n$ be a graded ring. A **graded left A-module** is a left A-module M together with a family $\{M_n\}_{n\geqslant 0}$ of subgroups of M such that*

(i) $M = \bigoplus_{n=0}^{\infty} M_n$ *as an abelian group*

(ii) $A_m M_n \subseteq M_{m+n}$ *for all $m, n \geqslant 0$.*

*Elements of M_n are called **homogeneous elements of degree** n.*

Remark. *Thus each M_n is an A_0-module.*

Definition 2.6. *Let A be a graded ring and M, N graded A-modules.*

1. *A **graded submodule** of M is a submodule N of M such that $N = \bigoplus_{n\geq 0} (N \cap M_n)$ in* **Ab**.

2. *then the $M/N = \bigoplus_{n\geq 0} M_n/(N \cap M_n)$ is well-defined, called the **quotient graded module** of M by N.*

3. *A **graded A-homomorphism** from M to N is an A-module homomorphism $f : M \to N$ such that $f(M_n) \subseteq N_n$ for all $n \geqslant 0$.*

§2.3 Graded algebra

Definition 2.7. *Let k be a commutative ring (usually a field). A **graded k-algebra** is k-algebra A such that*

(i) $A = \bigoplus_{n\geq 0} A_n$ *is a graded ring.*

(ii) the structure morphism $\varphi : k \to A$ with $\mathrm{Im}(\varphi) \subset A_0$ (scalar action preserves degrees).

Remark. *In this case, A_0 is naturally a k-algebra, and A may be viewed as a graded A_0-algebra with base ring A_0 instead of k. In many situations one further assumes that the base ring $k = A_0$..*

We give a graded ring of particular importance in commutative algebra.

Definition 2.8. *Let k be commutative ring (usually a field). A **standard graded k-algebra** is a graded k-algebra $A = \bigoplus_{n\geq 0} A_n$ such that A is generated by A_1 as a k-algebra.*

§3 Filtrations

Definition 3.1. *Let A be a ring. A descending* ***filtration*** *of A one means a sequence of subgroups*

$$A = A_0 \supset A_1 \supset A_2 \supset \cdots \supset A_n \supset \cdots$$

such that $A_m A_n \subseteq A_{m+n}$ for all $m, n \geqslant 0$. (thus each A_n is a two-sided ideal of A).

If $\mathfrak{a}$ is an ideal of A, then the $\{\mathfrak{a}^n\}$ is called the ***$\mathfrak{a}$-adic filtration****.*

Definition 3.2. *Let A be a filtered ring with descending filtration $\mathcal{F} = \{A_i\}$. A left A-module M is called a* ***filtered left A-module*** *if it has a sequence of subgroups (hence submodules) $\{M_i\}$ such that*

(i) $M = M_0 \supset M_1 \supset M_2 \supset \cdots \supset M_n \supset \cdots$

(ii) $A_i \cdot M_j \subseteq M_{i+j}$ *for all i, j.*

The $\{M_i\}$ is called a ***filtration*** *of M with respect to $\mathcal{F}$.*

Similarly, $\mathfrak{a}$-adic filtration of M is the filtration $\{\mathfrak{a}^n M\}$.

Definition 3.3. *Let A be a ring, $\mathfrak{a}$ an ideal and M an A-module with a filtration $\{M_n\}$.*

1. *We say that it is an* ***$\mathfrak{a}$-filtration*** *if $\mathfrak{a}M_n \subset M_{n+1}$ for all n.*
2. *We say that an $\mathfrak{a}$-filtration is* ***$\mathfrak{a}$-stable*** *if we have $\mathfrak{a}M_n = M_{n+1}$ for all n sufficiently large.*

Lemma 3.4. *If $\{M_n\}, \{M'_n\}$ are stable $\mathfrak{a}$-filtrations of M, then they have bounded difference: that is, there exists an integer n_0 such that $M_{n+n_0} \subseteq M'_n$ and $M'_{n+n_0} \subseteq M_n$ for all $n \geqslant 0$.*

Proof. Enough to take $M'_n = \mathfrak{a}^n M$. Since $\mathfrak{a}M_n \subseteq M_{n+1}$ for all n, we have $\mathfrak{a}^n M \subseteq M_n$; also $\mathfrak{a}M_n = M_{n+1}$ for all $n \geqslant n_0$ say, hence $M_{n+n_0} = \mathfrak{a}^n M_{n_0} \subseteq \mathfrak{a}^n M$. □

Lemma 3.5. *Let A be a Noetherian ring, M a finitely-generated A-module, (M_n) an $\mathfrak{a}$-filtration of M. Then the following are equivalent:*

1. *$M^* = \bigoplus_{n \geq 0} M_n$ is a finitely-generated A^*-module;*
2. *The filtration (M_n) is stable.*

Proof. Each M_n is finitely generated, hence so is each $Q_n = \oplus_{r=0}^n M_r$: this is a subgroup of M^* but not (in general) an A^*-submodule. However, it generates one in M^*, namely

$$M_n^* = M_0 \oplus \cdots \oplus M_n \oplus \mathfrak{a}M_n \oplus \mathfrak{a}^2 M_n \oplus \cdots \oplus \mathfrak{a}^r M_n \oplus \cdots$$

Since Q_n is finitely generated as an A-module, M_n^* is finitely generated as an A^*-module. The M_n^* form an ascending chain, whose union is M^*. Since A^* is Noetherian, M^* is finitely generated as an A^*-module $\Leftrightarrow$ the chain stops, i.e., $M^* = M_{n_0}^*$ for some $n_0 \Leftrightarrow M_{n_0+r} = \mathfrak{a}^r M_{n_0}$ for all $r \geqslant 0 \Leftrightarrow$ the filtration is stable. □

Proposition 3.6 (Artin-Rees)**.** *Let A be a Noetherian ring, $\mathfrak{a}$ an ideal, M a finitely generated A-module and $\{M_n\}$ a stable $\mathfrak{a}$-filtration. If M' is a submodule of M, then $\{M' \cap M_n\}$ is a stable $\mathfrak{a}$-filtration of M'.*

Proof. We have $\mathfrak{a}\,(M' \cap M_n) \subseteq \mathfrak{a}M' \cap \mathfrak{a}M_n \subseteq M' \cap M_{n+1}$, hence $(M' \cap M_n)$ is an α-filtration. Hence it defines a graded A^*-module which is a submodule of M^* and therefore finitely generated. □

Corollary 3.7. *There exists an integer s such that for all integers $n \geq s$ we have*

$$\mathfrak{a}^n M \cap M' = \mathfrak{a}^{n-s}\left(\mathfrak{a}^s M \cap M'\right)$$

Corollary 3.8. *Let A be a Noetherian ring, a an ideal, M a finitely-generated A-module and M' a submodule of M. Then the $\mathfrak{a}$-topology of M' coincides with the topology induced by the $\mathfrak{a}$-topology of M.*

§4 Associated graded ring

Definition 4.1. *Let A be a ring with filtration $\mathcal{F}$, and M an filtered A-module with respect to $\mathcal{F}$.*

1. *We define **associated graded ring** of A with respect to $\mathcal{F}$*

$$\mathrm{gr}_{\mathcal{F}}(A) = \bigoplus_{n=0}^{\infty} A_n/A_{n+1}$$

2. *We define **associated graded module** of A with respect to $\mathcal{F}$*

$$\mathrm{gr}(M) = \bigoplus_{n=0}^{\infty} M_n/M_{n+1}$$

then $\mathrm{gr}(M)$ is a graded $\mathrm{gr}_{\mathcal{F}}(A)$-module.

Proposition 4.2. *Let A be a Noetherian ring and $\mathfrak{a}$ an ideal. Then*

1. *$\mathrm{gr}_{\mathfrak{a}}(A)$ is Noetherian.*
2. *If M is a finitely generated A-module with a stable $\mathfrak{a}$-filtration, then $\mathrm{gr}(M)$ is a finitely generated graded $\mathrm{gr}_{\mathfrak{a}}(A)$-module.*

Chapter X

Dimension

§1 Hilbert Functions

Definition 1.1. *Let $\mathcal{A}$ be a abelian category. A* ***additive function*** $\lambda : \mathcal{A} \to$ **Ab** *is a covariant functor such that for every short exact sequence*

$$0 \to M' \to M \to M'' \to 0$$

in $\mathcal{A}$, we have

$$\lambda(M) = \lambda(M') + \lambda(M'')$$

In the following, $\lambda(N)$ is a finitely generated abelian group and we always view $\lambda(N)$ as the rank of $\lambda(N)$.

Definition 1.2. *Let $A = \bigoplus_{n \geq 0} A_n$ be a commutative Noetherian graded ring and $M = \bigoplus_{n \geq 0} M_n$ a finitely generated graded A-module (It follows that each M_n is a finitely generated A_0-module with a additive function λ).*

1. *The* ***Hilbert function*** *of M with respect to λ is the function defined by*

$$h_M^\lambda(n) := \lambda(M_n)$$

2. *the* ***Hilbert-Poincare series*** *of M*

$$P_M^\lambda(t) := \sum_{n=0}^{\infty} h_M^\lambda(n) t^n$$

Remark. *If A_0 is Artinian ring, $\lambda(N) = \ell_{A_0}(N)$. If A_0 is a field, $\lambda(N) = \dim_{A_0}(N)$.*

Theorem 1.3 (Hilbert-Serre)**.** *The Poincare series $P_M^\lambda(t)$ is a rational function of the form*

$$\frac{f(t)}{\prod_{i=1}^{d}(1 - t^{k_i})}$$

where $f(t) \in \mathbb{Z}[t]$ *and* $k_1, \ldots, k_d$ *are the degrees of the homogeneous generators of* A *over* A_0. *The order of the pole of* $P_M^\lambda(t)$ *at* $t = 1$ *is called the* ***dimension*** *of* M *with respect to* λ*, denoted by* dim M.

Proof. By induction on s, the number of generators of R over R_0. Start with $s = 0$; this means that $R_n = 0$ for all $n > 0$, so that $R = R_0$ and M is a finitelygenerated R_0 module, hence $M_n = 0$ for all large n. Thus $P_M^\lambda(t)$ is a polynomial in this case.

Now suppose $s > 0$ and the theorem true for $s - 1$. Multiplication by x_s is an R-module homomorphism of M_n into M_{n+k_s}, hence it gives an exact sequence, say

$$0 \to K_n \to M_n \xrightarrow{x_s} M_{n+k_s} \to L_{n+k_s} \to 0.$$

Let $K = \oplus_n K_n, L = \oplus_n L_n$; these are both finitely-generated A-modules (because K is a submodule and L a quotient module of M), and both are annihilated by x_s, hence they are $A_0\left[x_1, \ldots, x_{s-1}\right]$-modules. Applying λ to (1) we have, by (2.11)

$$\lambda\left(K_n\right) - \lambda\left(M_n\right) + \lambda\left(M_{n+k_s}\right) - \lambda\left(L_{n+k_s}\right) = 0;$$

multiplying by t^{n+k_s} and summing with respect to n we get

$$\left(1 - t^{k_s}\right) P(M, t) = P(L, t) - t^{k_s} P(K, t) + g(t)$$

where $g(t)$ is a polynomial. Applying the inductive hypothesis the result now follows. □

Corollary 1.4 (Existence of the Hilbert polynomial)**.** *Then there exists a polynomial* $H(t) \in \mathbb{Q}[t]$ *of degree* dim M *such that* $H(n) = h_M^\lambda(n)$ *for all sufficiently large integers* n.

This polynomial is called the ***Hilbert polynomial*** *of* M *with respect to* λ.

Corollary 1.5. *There exists a polynomial* $\chi(t, A, M) \in \mathbb{Q}[t]$ *such that*

$$\sum_{i=0}^{n} \lambda\left(M_i\right) = \chi(n)$$

for all sufficiently large integers n.

Definition 1.6. *The* ***multiplicity*** *of* M *with respect to* λ *is defined to be*

$$e(M, \lambda) := d! \cdot a_{d(M)}$$

where $a_{d(M)}$ *is the leading coefficient of the Hilbert polynomial* $\chi(t)$.

§2 Krull Dimension

In this section, all rings A are commutative Noetherian ring with identity and modules M are finitely generated A-module. Although some definitions and results hold in greater generality, the most of useful properties stay in this case.

Definition 2.1. *Let A-module M.*

1. *The **Krull dimension** of A is defined to be the supremum of the lengths n of all chains of prime ideals*
$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$
say $\dim(A) = n$ *if the supremum is attained, and* $\dim(A) = \infty$ *otherwise.*

2. *The **Krull dimension** of A-module M is defined to be the dimension of the quotient ring $A/\operatorname{Ann}(M)$, that is,*
$$\dim(M) := \dim\left(A/\operatorname{Ann}(M)\right)$$

Proposition 2.2. .

1. $\dim A = \dim \operatorname{Spec} A$

2. *If A is a local ring with maximal ideal $\mathfrak{m}$, then* $\dim A = \sup\left\{n : \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathfrak{m}\right\}$

3. *if S is a multiplicatively closed subset of A, then* $\dim S^{-1}A \leq \dim A$

Definition 2.3. *Let A be a commutative ring and $\mathfrak{p}$ a prime ideal of A. The **height** of $\mathfrak{p}$ is defined to be*
$$\operatorname{ht}(\mathfrak{p}) := \sup\left\{n : \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathfrak{p}\right\}$$
Thus $\dim A = \sup_{\mathfrak{p} \in \operatorname{Spec} A} \operatorname{ht}(\mathfrak{p})$.

Theorem 2.4 (Krull Principal Ideal Theorem)**.** *Let A be a Noetherian commutative ring and a ideal $\mathfrak{a} = (f_1, \ldots, f_r)$. If $\mathfrak{p}$ is a minimal element of* $\{\mathfrak{q} \in \operatorname{Spec} A : \mathfrak{a} \subset \mathfrak{q}\}$. *Then* $\operatorname{ht}(\mathfrak{p}) \leq r$.

Chapter XI

§1

Definition 1.1. *The space X is called **Noetherian** if it satisfies the descending chain condition for closed subsets, i.e., every descending chain of closed subsets eventually stabilizes.*

Definition 1.2. *The space X is called **irreducible** if it cannot be expressed as the union of two proper closed subsets. The subset Y of X is called **irreducible** if it is irreducible as a subspace of X, and empty set is irreducible.*

Remark. *It is easy to see that X is irreducible if and only if*

1. *Any two non-empty open subsets of X intersect. Any two non-empty closed subsets of X intersect.*
2. *Every non-empty open subset of X is dense in X. Every non-empty closed subset of X has empty interior.*

§2 Spectrum of a Ring

Definition 2.1. *Let A be a commutative ring with identity. The **spectrum** of A, denoted by*

$$\operatorname{Spec} A := \{\mathfrak{p} : \mathfrak{p} \textit{ is a prime ideal of } A\}$$

*, is the set of all prime ideals of A. If $\mathfrak{a}$ is an ideal of A, then the **vanishing (zeros) set** of $\mathfrak{a}$ is defined to be*

$$V(\mathfrak{a}) := \{\mathfrak{p} \in \operatorname{Spec} A : \mathfrak{a} \subset \mathfrak{p}\} \subset \operatorname{Spec} A$$

*The **Zariski topology** on* $\operatorname{Spec} A$ *is defined by taking the collection of all vanishing sets $V(\mathfrak{a})$ as the closed sets.*

Definition 2.2. *Defing a sheaf $\mathcal{O}$ on $X =$* $\operatorname{Spec} A$*, for each open set $U \subseteq X$, we define $\mathcal{O}(X)$ be the set of functions*

$$s : U \to \coprod_{\mathfrak{p} \in U} A_{\mathfrak{p}}, \quad \textit{or simplified } (s_{\mathfrak{p}})_{\mathfrak{p} \in U}$$

such that $s(\mathfrak{p}) \in A_{\mathfrak{p}}$ *for all* $\mathfrak{p} \in U$*, and such that* s *is locally a quotient of elements of* A*, i.e., for each* $\mathfrak{p} \in U$*, there exists an open neighborhood* $p \in V \subseteq U$ *and elements* $a, f \in A$ *such that* $f \notin \mathfrak{q}$ *for all* $\mathfrak{q} \in V$ *and* $s|_V = a/f$.

Definition 2.3. *Let* A *be a commutative ring with identity. The* ***spectrum*** *of* A *is the pair consisting of the topological space* Spec A *and the sheaf of rings* $\mathcal{O}$ *defined above.*

Definition 2.4. *Let* $f \in A$*, the* ***distinguished open set*** *of* f *is defined to be*

$$D(f) := \{\mathfrak{p} \in \operatorname{Spec} A : f \notin \mathfrak{p}\} = \operatorname{Spec} A - V(f)$$